PROFORMA FOR THE APPROVAL PROJECT PROPOSAL

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1. **Name of the Student**:
   Raj Ramji Verma And Suraj Premchandra Singh

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2. **Title of the Project :**
   BROK FREE ANDROID APP

3. **Name of the Guide** :
   Prof. Hina Mahmood

4. **Teaching experience of the Guide** : 10 Years

5. **2=890** ?

YES
NO

Signature of the Student   Signature of the Guide

Date: …………………   Date: …………………….

Signature of the Coordinator

Date: …………………

BROK FREE APP

A Project Report

Submitted in partial fulfillment of the

Requirements for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

By

Raj Ramji Verma

&

Suraj Premchandra Singh

Under the esteemed guidance of

Ms. Hina Mahmood

Asst. Professor(I.T. Dept.)

DEPARTMENT OF INFORMATION TECHNOLOGY

RIZVI COLLEGE OF ARTS, SCIENCE & COMMERCE

(Affiliated to University of Mumbai)

MUMBAI, 400050

MAHARASHTRA

2021-2022

RIZVI COLLEGE OF ARTS, SCIENCE & COMMERCE

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MUMBAI-MAHARASHTRA-400050

DEPARTMENT OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that the project entitled, **"BROK FREE APP "**, is bonafied work of **Raj Verma** bearing Seat.No: (**14**) **& Suraj Singh** bearing Seat.No: (**03**) submitted in partial fulfillment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY from University of Mumbai. (12, times new roman, justified)

Internal Guide | **Coordinator**

External Examiner

Date: | **College Seal**

Abstract

BrokFree android application is a platform that helps to buy sell and rent your property. It gives a platform that without give any broker charge and help the sell, buy and rent your property. It has many features that help to best deal for your property.As we know when we find property, we need broker. Who helps as tocontact to seller and requests some amount of commission on it So,thisapp helps buyer to search for a property and contact the seller without a need of broker which saves the commission fee

ACKNOWLEDGEMENT

The success and final outcomes of this project required a lot of guidance and assistance from many people and I am extremely privileged to have got this all along the completion of my project. All that I have done is only due to such supervision and assistance and we would not forget to thank them.

We would like to express our special thanks of gratitude to our guide and supervisor **Prof. Hina Mahmood** who gave us the golden opportunity to do this wonderful project on **Brok Free App**, which also helped us in doing a lot of research and we came to know about so many new things about android. We are really thankful to her.

Date :-

Name Of The Students:
Raj Ramji Verma
Suraj Premchandra Singh

DECLARATION

I here by declare that the project entitled, "**BrokFree (Android App)**" done at **place where the project is done**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfillment of the requirements for the award of degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY) to be submitted

as final semester project as part of our curriculum.

Name and Signature of the Student

TABLE OF CONTENT

Synopsis

Introduction

BrokFree android application is a platform that helps to buy sell and rent your property. It gives a platform that without give any broker charge and help the sell, buy and rent your property. It has many features that help to best deal for your property.

It helps to classify diffident category for property and manage the best deal like commercial, residential, flat, bang-lo, society etc. So much Category help to manage system very well manner and you well ableto buy slum area.in this system work on only seller and buyer no onecan involve that system. Buyer can communicate direct to seller thorough this application and you safely communicant withbuyer/seller without any broker.

To improve and help buying property very well manner using price predictor that manage price using AI (Artificial Intelligence) and ML (machine lingering) algorithm that give to price prediction. In this algorithm we get the specific data like area of property, build area, nearby facilities, location, category etc. And also, to detect the location and around 2 km radius to find facility that help to buy bestproperty efficient way. The algorithm also gets the system serve and government regulation data for property that manage that propertyprice. To show the buyer price and our ML algorithm price to computerization.

Purpose

Our application that helps to buy, sell and rent your property without any broker charge. You can buy some different category that help to buy best deal. You also deal with some slum area. to manta best deal pricepredictor that help to predict different area. And you well be able check the incoming fasciitis like metro road, hotels, school, colleges etc. User can search by their locality.

Scope

To making good decision to buying your property and you pay the best price for your property without any broker charge. You save the 3 to 4% of money for broker charge. And some time many people was scam with broker that help to protect. And you will directly buy property thorough builder that make complicit broker free and reador buildup category.

Existing System

99acers: -

In this available play store and it provide website view also to direct communication thorough web site and user can Buy, Sell, Rent, properties with loan system. in this app commercial and residential properties are available. it also supports big industrial projects also here safety features are available and it also provides real time map. And even upcoming event that provide well manner but it does not have slum area projects and properties. here user cannot bargain.

Remove Brokers: in this app user can only sale and buy properties there are safety features are available. it does not show prize variation. And it not gives to rent feature. And it does not support the lone system. Upcoming event normally not show and not communicate the buyer directly and pay some charges to show best dial .so many advertisements in this apothem management not be proper **Advantage**

- User can buy, sale and rent properties easily.
- User can direct contact to owner or builder.
- Users can buy properties located in slum areas..
- The data will be keep updating.
- Property prize according to govt. rules and regulations also will beshown.
- The prize of property will be shown on the basis of owner's demand,govt. rules and regulations and nearby places and facilities.
- User can search by their locality.

Chapter 1

Existing System: -

1) Nobroker Brokers:

In this available play store and it provide website view also to direct communication thorough web site and user can Buy, Sell, Rent, properties with loan system. in this app commercial and residential properties are available. it also supports big industrial projects also here safety features are available and it also provides real time map. And even upcoming event that provide well manner but it does not have slum area projects and properties. here user cannot bargain.

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Objective

Our stem that gets save time. reduce the broker charge and tofeting the best prize in comm paring the seller prize, government issues prize, and disarming commingle feature as well as current feature. And get to making flume area that sell in our system

Proposed System

To making good decision to buying your property and you pay the best price for your property without any broker charge. You save the 3 to 4% of money for broker charge. And some time many people was scam with broker that help to protect. And you will director buy

property thorough builder that makes complicit broker free and read or buildup category.

Use the new system that help to price predication and caking the property surrounding area or facility, and upcoming develop event and to checking some slam areas.

Scope

To making good decision to buying your property and you pay thebest price for your property without any broker charge. You save the3 to 4% of money for broker charge. And some time many people was scam with broker that help to protect. And you will director buy property thorough builder that make complicit broker free and readorbuildup category.

Applicability

- User can experience broker free breakage to find their property
- The user can invest in any kind of areas like rural, urban,developed, under development and slum areas also.

- Buyers can compare price of any property location according togovernment aspects, seller expectations and nearby facilities.
- Users can check available nearby facilities with distance from therespective location.
- Future project details and updates also will be available so that itbecomes easy to find required plot or property.
- Buyers can directly communicate to sellers so that discuss aboutthe deal.
- Project industries are also involved so that they can easily selltheir flats in bulk and buyers can get a better deal.
- It will also reduce the scams of the real estate industry and gives afearless environment for real estate deals.

Achievement

- To provide broker free services.
- To give golden chance for Buying property.
- To invest all type of property.
- Direct communication service.
- To comparing to prize factor locality.
- Provide best serve for selling property (to know to area).
- To get best price.
- Free advertise property.
- To filter the best deals

Organization report

First major step to get property details and add in our application. Seller Lille House detail, acclivity feature, commingle feature

And other hand industry Ravel fill same data and some extra like past infrastructural, property deal, eagle data, and incoming infrastructural and ether Dustin Mantegna system details Sec and step buyer can see all post and direct communicate seller

Chapter 2

AnalysisRequirement gathering Available Technologies

Java

Since the introduction of Android in 2008, this object-oriented programming language has been the popular and official language Android mobile app development. Extremely versatile language, Java helps keep your app flexible, modular, and extensible. Java is easy to handle and many opensource libraries are madeavailable for users to choose from.

Flutter

Flutter is the new trending cross-platform mobile application development technology in town. It uses “Dart” as a programming language instead of JavaScript which facilitates rapid and effective analysis, fabricates AIs, includes highlights and fixes bugs in milliseconds. The open-source cross-platform SDK by Google extends a wide range of plugins backed by Google and allows mobile apps to be built for both Android and Apple DOS platforms.

Firebase

When it comes to the data-synchronization and offline data modifying – it can be a number one local solution to go with. Firebase supports a real-time NoSQL database for React Native applications and it is capable enough to fulfil the MVC requirements for them. Another strong advantage of it is a crossplatform API with minimumrequirements for setup. Moreover, it can be easily directly accessedas a real-time database from a mobile device.

kotlin kotlin is a statically typed programming language that is used for developing modern Android applications today. Kotlin is said to be theadvanced version of Java. For Android developers, this language is amodern answer to obsolete Java.

kotlin is a language that has influenced some other programming languages such as Callas, Gosh, Groovy, Java, etc.

Advantages with proposed system

Advantages:

- Java is a high-level programming language.
- Java supports portability feature.

- Java provides Automatic Garbage Collection.
- Java supports multi-skilling
- Java provides an efficient memory allocation strategy

Disadvantages:

- Java is slow and has a poor performance
- Java requires significant memory space Verbose and Complex codes.

Flutter

Advantages:

- Same UI and Business Logic in All Platforms.
- Reduced Code Development Time.
- Increased Time-to-Market Speed.
- Similar to Native App Performance.

Disadvantages:

- The apps made with Flutter tend to be weighty ones.
- Flutter-based apps are not supported by browsers as of now. Thismeans no web apps.
- While Flutter is popular, it has not been around long enough to have a huge resource base. Therefore, your team will need to write a lot of stuff from scratch.

kotlin

Advantages:

- Readability
- Null-Safe
- Using Begetters and Setters

Disadvantages

Limited sources to learn.

- Fluctuating compilation speed.

Firebase

Advantages:

- Reliable & Extensive Databases.
- Fast & Safe Hosting.
- Provides A Free Start to Newbies.
- Firebase Cloud Messaging for Cross-Platform.

Disadvantages:

- very limited querying and indexing.
- no aggregation.
- can’t query or list users or stored file

Feasibility study

The objective behind the feasibility study is to create the reasons for developing the software that is acceptable to users, flexible to change and conformable to established standards.

Feasibility Study can be considered as preliminary investigation that helps the management to take decision about whether study of system should be feasible for development or not.

It identifies the possibility of improving an existing system, developing a new system, and produce refined estimates for further development of system. it is used to obtain the outline of the problem and decide whether feasible or appropriate solution exists or not.

The main objective of a feasibility study is to acquire problem scope instead of solving the problem. the output of a feasibility study is a formal system proposal act as decision document which includes the complete nature and scope ofthe proposed system.

Types of Feasibility:

Economic Feasibility

- It is evaluating the effectiveness of candidate system by usingcost/benefit analysis method.
- It demonstrates the net benefit from the candidate system interms of benefits and costs to the organization.
- The main aim of Economic Feasibility Analysis (EFS) is to estimate the economic requirements of candidate system before investments funds are committed to proposal.

- It prefers the alternative which will maximize the net worth of organization by earliest and highest return of funds along with lowest level of risk involved in developing the candidate system. requirements within the time and budget.

Technical Feasibility

- It investigates the technical feasibility of each implementationalternative.
- It analyzes and determines whether the solution can be supportedby existing technology or not.
- The analyst determines whether current technical resources beupgraded or added it that fulfill the new requirements.
- It ensures that the candidate system provides appropriate responses to what extent it can support the technical enhancement.

Schedule Feasibility

- It ensures that the project should be completed within given time constraint or schedule.
- It also verifies and validates whether the deadlines of project arereasonable or not.

Behavioral Feasibility

- It evaluates and estimates the user attitude or behavior towardsthe development of new system.
- It helps in determining if the system requires special effort to educate, retrain, transfer, and changes in employee's job statuson new ways of conducting business.

Operational Feasibility

- It determines whether the system is operating effectively once itis developed and implemented.
- It ensures that the management should support the proposed system and its working feasible in the current organizational environment.
- It analyzes whether the users will be affected and they accept the modified or new business methods that affect the possible system benefits.
- It also ensures that the computer resources and network architecture of candidate system are workable.

Requirement specification

I am using android studio to building our app there are design back and using java code. And front-end design by XML and suing database system Go ogles Firebase Database, Java and some git hub library Apache poi for creating excel file to management data

And some tenser flow library to generate machine learning code to genre-at environment to run the algorithm and get the system data and predict the price and some other library to design system user interface.

Also use Google col lab to run machine learning algorithm it testing purpose

Hardware Requirements

For working application Only required smart phone no another physical hardware required. The system run in android 4 or above andsystem also contain more than 2Gb ram ,200mb internal storage space, Wi-Fi or internet connection, internal Geo-location to find area.When making the application we required more than 4GB ram ,75GBinternal storage that mange package library and some data 2.3 GHz Pentium processor etc.

Not required physical hardware to making this application.

Software Requirements

Android studio is the platform to making android application.it work on both java and kotlin language our application making on java language in system built in virtual mobile is available for testing.

The system has support for built in database like Google Firebase that also support java and kotlin its real time database, Firestore and storage that contain j son format and data store in tree like structure to manage the data batter for table format.

Android environment also help Google library support that manage many complex things in well manner that contain ml algorithm face detection, bar code scanner and many thinks it also contain external algorithm using tenser flow library it supports java and kotlin language.

For user interface use the android studio built in XML system to making fast and simple inter face and it easy to marge both language

Justification of selection of Technology

- **Android studio:**

The system as many feature that help very Mach to develop application there are inbuilt function and library support to manage thinks

- **User interface (front end):**

The system support XML language that helps to making user interfaceand it easy to integrated the system language like java mandolin language .it also making j son language in background manage wholesystem. It is best for making UI because the XML is easy language andthe more function for other language.to making best UI fast way.

- **Coding language (back end) (java):**

The system supports multiple language like java, Hotlink etc. Our application making in java language because the java developer is big community to support and java is also high-level language to helpmany platforms structure like system integrate by supports etc. to easy way to manage all thing in one language.

Database (Firebase):

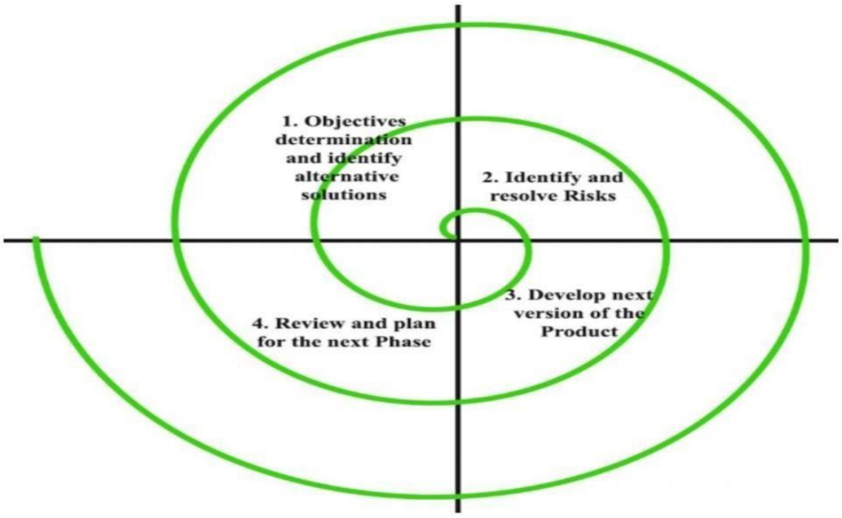
The system also contain build in database to manage thinks the most popular database and very fast, efficient, real-time database is Firebase. The Firebase database is making Google that tasting purposefree of cost as well as the key feature is that it stores a in tree like structure in j son format it very efficient other table like structure like SQL, it accesses directly to cloud there for it work faster

Chapter 3

Detailed life cycle of the project (Logical Design): -

Spiral Model

The spiral model, initially proposed by Beerbohm, is an evolutionary software process model that couples the iterative feature of prototyping with the controlled and systematic aspects of the linear sequential model. It implements the potential for rapid development of new versions of the software. Using the spiral model, the software is developed in a series of incremental releases. During the early iterations, the additional release may be a paper model or prototype. During later iterations, more and more complete versions of the engineered system are produced.



Objective setting:

Each cycle in the spiral starts with the identification of purpose forthat cycle, the various alternatives that are possible for achieving thetargets, and the constraints that exists.

Risk Assessment and reduction:

The next phase in the cycle is to calculate these various alternatives based on the goals and constraints. The focus of evaluation in this stage is located on the risk perception for the project.

Development and validation:

The next phase is to develop strategies that resolve uncertainties and risks. This process may include activities such as bench marking,simulation, and prototyping.

Planning:

Finally, the next step is planned. The project is reviewed, and a choice made whether to continue with a further period of the spiral. Ifit is determined to keep, plans are drawn up for the next step of the project.

The development phase depends on the remaining risks. For example, if performance or user-interface risks are treated more essential than the program development risks, the next phase may be an evolutionary development that includes developing a more detailed prototype for solving the risks.

The risk-driven feature of the spiral model allows it to accommodate any mixture of a specification-oriented, prototype-oriented, simulation-oriented, or another type of approach. An essential element of the model is that each period of the spiral is completed by a review that includes all the products developed during that cycle, including plans for the next cycle. The spiral model works for development as well as enhancement projects.

Advantages of Spiral Model:

- Software is produced early in the software life cycle.
- Risk handling is one of important advantages of the Spiral model, it is best development model to follow due to the risk analysis andrisk handling at every phase.
- Flexibility in requirements. In this model, we can easily change requirements at later phases and can be incorporated accurately. Also, additional Functionality can be added at a later date.
- It is good for large and complex projects.
- It is good for customer satisfaction. We can involve customers in the development of products at early phase of the softwaredevelopment. Also, software is produced early in the software lifecycle.
- Strong approval and documentation control.
- It is suitable for high-risk projects, where business needs may be unstable. A highly customized product can be developed using this.

Disadvantages of Spiral Model:

- It is not suitable for small projects as it is expensive.
- It is much more complex than other SDLC models. Process is complex.
- Too much dependable on Risk Analysis and requires highly specific expertise.
- Difficulty in time management. As the number of phases is unknown at the start of the project, so time estimation is very difficult.
- Spiral may go on indefinitely.
- End of the project may not be known early.
- It is not suitable for low-risk projects.
- May be hard to define objective, verifiable milestones. Large numbers of intermediate stages require excessivedocumentation.

Flow chart: -

A flowchart is a diagram that depicts a process, system or computer algorithm. They are widely used in multiple fields to document, study, plan, improve and communicate often complex processes in clear, easy-to-understand diagrams. Flowcharts, sometimes spelled as flow charts, use rectangles, ovals, diamonds and potentially numerous other shapes to define the type of step, along with connecting arrows to define flow and sequence.

Basic Flowchart Symbols and Notations:

- **Terminal:**

the flowchart, it is represented with the help of a circle for denoting the start and stop symbol. The symbol given below is usedto represent the terminal symbol.

- **Input/output Symbol:**

The input symbol is used to represent the input data, and the output symbol is used to display the output operation. The symbol given below is used for representing the Input/output symbol.

- **Decision Symbol:**

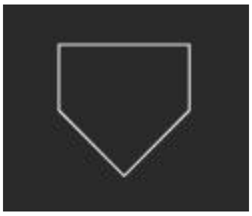
Diamond symbol is used for represents decision-making statements.The symbol given below is used to represent the decision symbol.

- **Flow lines:**

It represents the exact sequence in which instructions are executed. Arrows are used to represent the flow lines in a flowchart. The symbol given below is used for representing the flow lines:

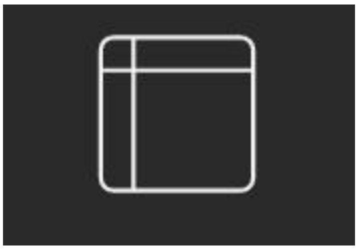
- **Off-Page Reference:**

This symbol contains a letter inside indicating that the flow continues on a matching symbol containing the same letter somewhere else on a different page. The symbol given below is usedto represent the off-page reference symbol.



- **Internal storage symbol:**

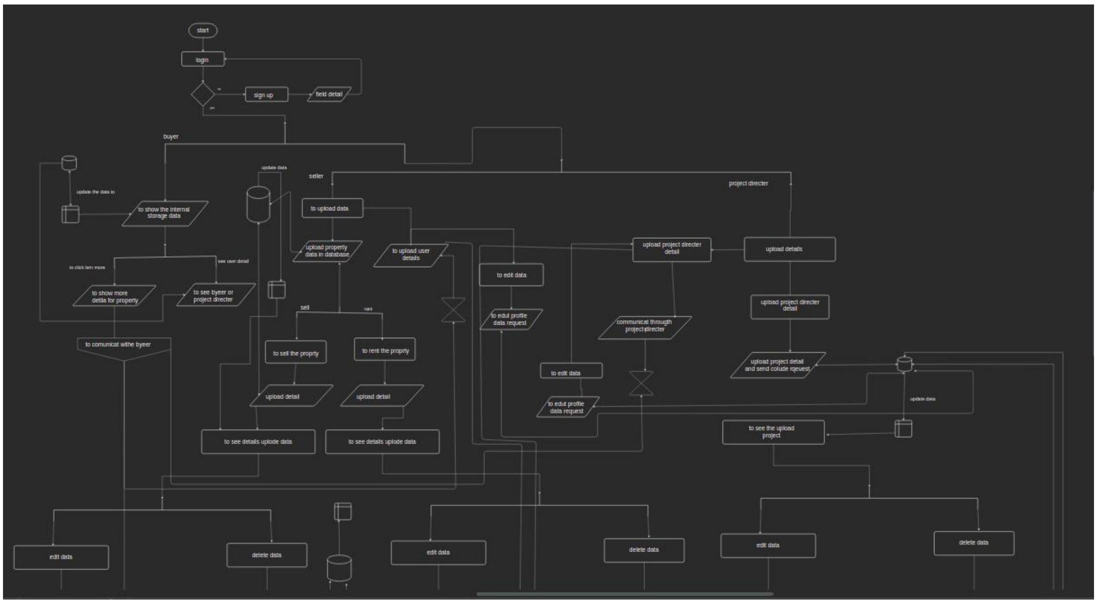
The symbol given below is used to represent the internal storage symbol.to use store data use in system

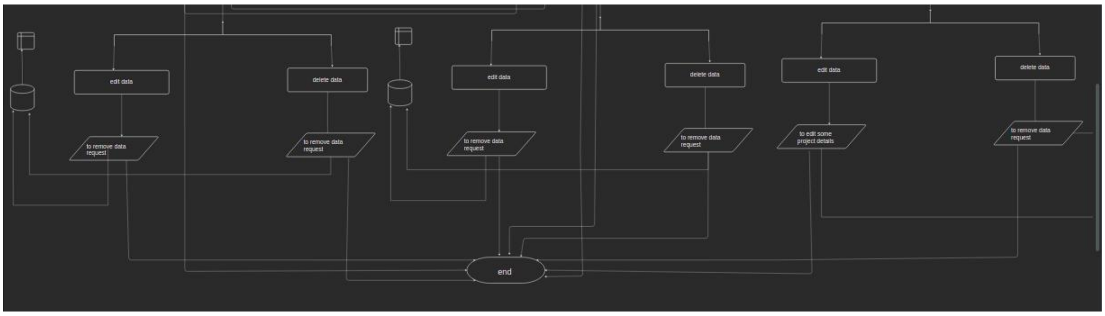


- **Cloud storage /database:**

The symbol given below is used to represent the database symbol. Alldata are set into cloud to get backup

Flow chart diagram





Class Diagram: -

Class diagram is a static diagram. It represents the static view of an application. Class diagram is not only used for visualizing, describing, and documenting different aspects of a system but also for constructing executable code of the software application.

Class diagram describes the attributes and operations of a classand also the constraints imposed on the system. The class diagramsare widely used in the modelling of object-oriented systems becausethey are the only UML diagrams, which can be mapped directly with object-oriented languages.

Basic Class Diagram Elements:

Class: -

- **Upper Section:**

The upper section encompasses the name of the class. A class is a representation of similar objects that shares the same relationships,attributes, operations, and semantics. Some of the following rules that should be taken into account while representing a class are givenbelow: Capitalize the initial letter of the class name.

Place the class name in the center of the upper section.A class name must be written in bold format.

The name of the abstract class should be written in italics format.

- **Middle Section:**

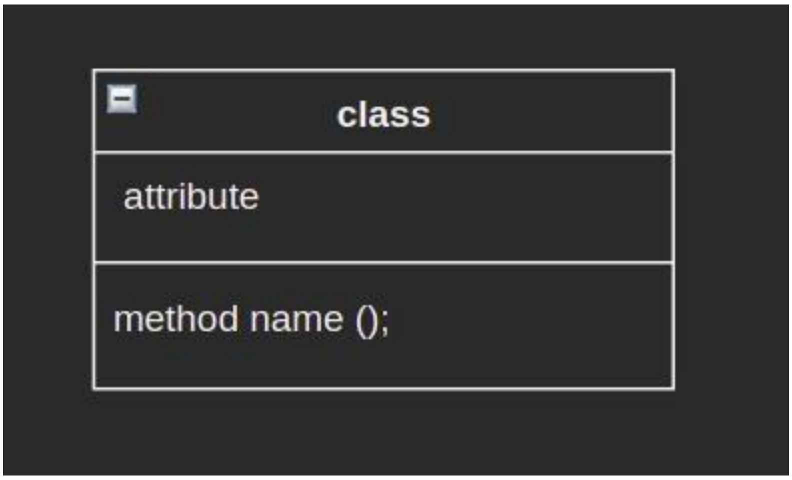
The middle section constitutes the attributes, which describe the quality of the class. The attributes have the following characteristics: The attributes are written along with its visibility factors, which arepublic (+), private (-), protected (#), and package (~).

The accessibility of an attribute class is illustrated by the visibility factors.

A meaningful name should be assigned to the attribute, which willexplain its usage inside the class.

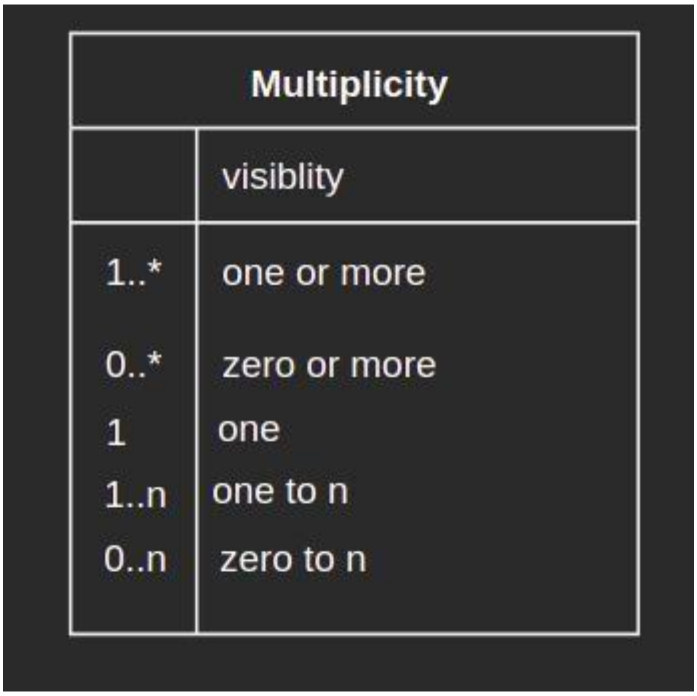
- **Lower Section:**

The lower section contains methods or operations. The methods are represented in the form of a list, where each methodis written in a single line. It demonstrates how a class interacts with data.



- **Multiplicity**:

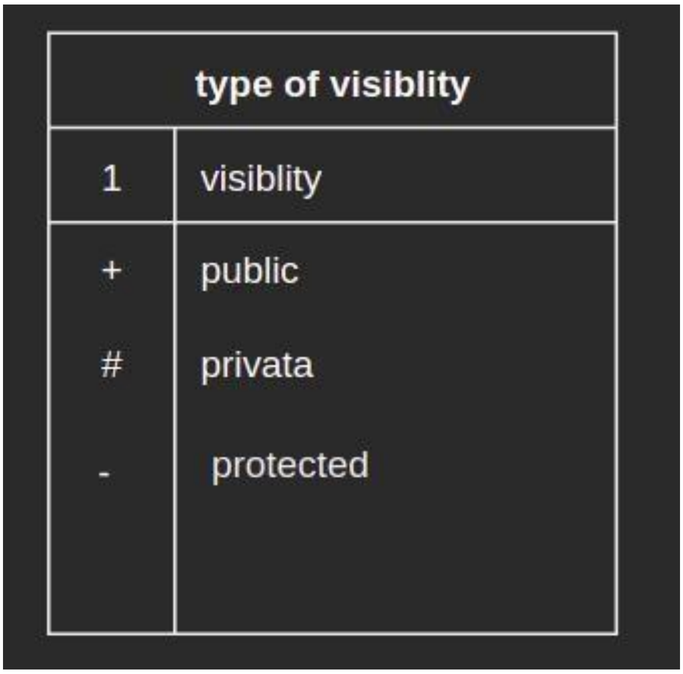
A multiplicity allows for statements about the number of objects thatare involved in an association. These symbols indicate the number ofinstances of one class linked to one instance of the other class.



Example

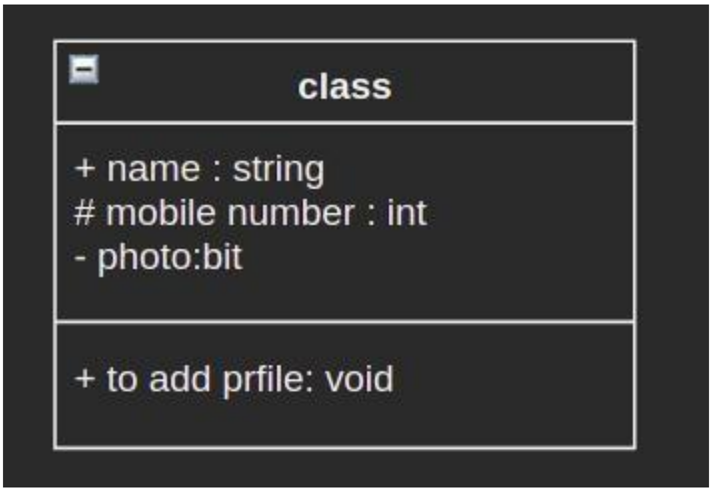
4. **Visibility** –

Use visibility markers to signify who can access the information contained within a class. Private visibility, denoted with a - sign, hides information from anything outsidethe class partition. Public visibility, denoted with a + sign, allows all other classes to view the marked information. Protected visibility, denoted with a # sign, allows child classes to access information they inherited from a parent class.



type of visiblity	
1	visiblity
+	public
#	privata
-	protected

example



Class diagram relationAssociation:

is a broad term that encompasses just about any logical connection or relationship between classes. For example, passenger and airline may be linked as above:

Directed Association:

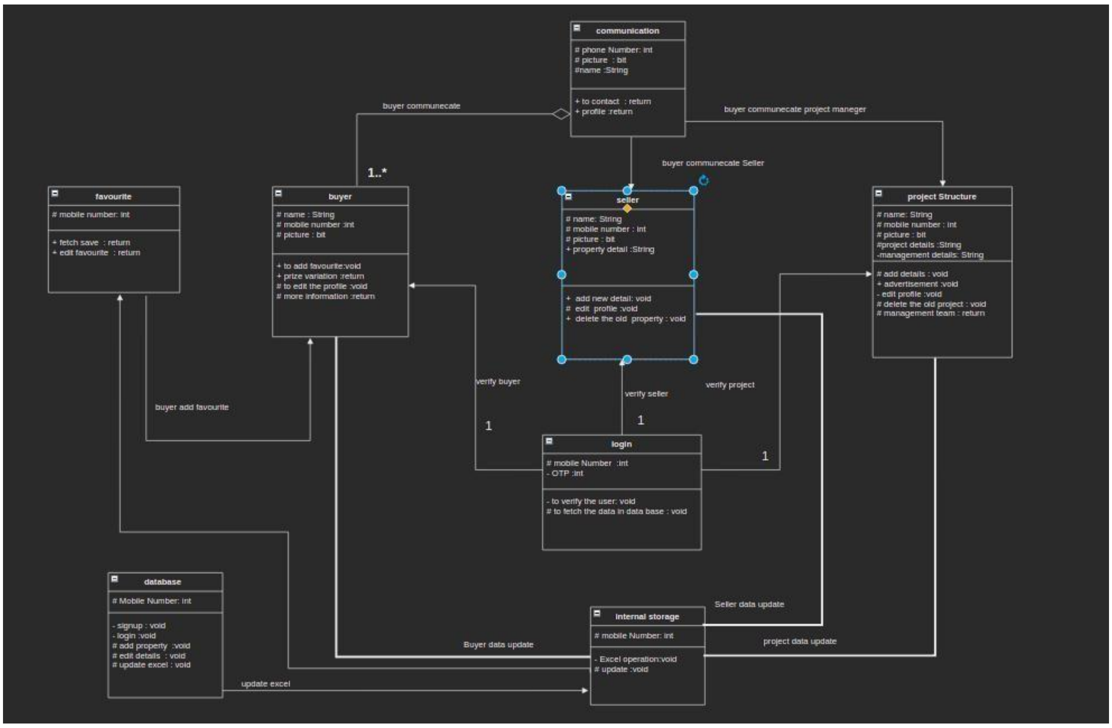
refers to a directional relationship represented by a line with an arrowhead. The arrowhead depicts a container-contained directionalflow.

Reflexive Association

This occurs when a class may have multiple functions or responsibilities. For example, a staff member working in an airport may be a pilot, aviation engineer, a ticket dispatcher, a guard, or a maintenance crew member. If the maintenance crew member is managed by the aviation engineer there could be a managed by relationship in two instances of the same class.

Multiplicity is the active logical association when the cardinal of a class in relation to another is being depicted. For example, one fleet may include multiple airplanes, while one commercial airplane may contain zero to many passengers. The notation 0..* In the diagram means “zero to many”

Class diagram



ER Diagram: -

ER Diagram stands for Entity Relationship Diagram, also known asERD is a diagram that displays the relationship of entity sets stored in a database. In other words, ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basicconcepts: entities, attributes and relationships.

an ERD can be useful for organizing data that can be represented by a relational structure, it can't sufficiently represent semi-structured or unstructured data. It's also unlikely to be helpful on its own in integrating data into a pre-exist information system.

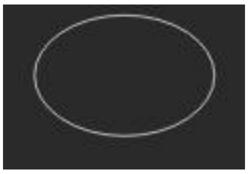
Basic ER Diagram Symbols and Notations:

Entity:

Entities are represented by rectangles. An entity is an object or concept about which you want to store information. A weak entity isan entity that must defined by a foreign key relationship with another entity as it cannot be uniquely identified by its own attributes alone.

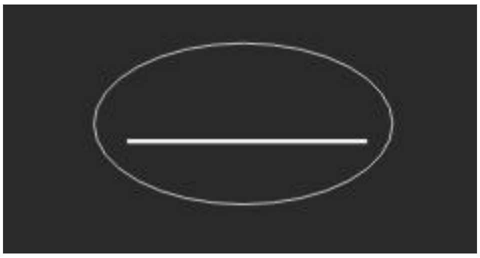
Attribute:

Attributes are the properties which define the entity type. For example, Roll No, Name, DOB, Age, Address, Mobile No are the attributes which defines entity type Student. In ER diagram, attributeis represented by an oval.



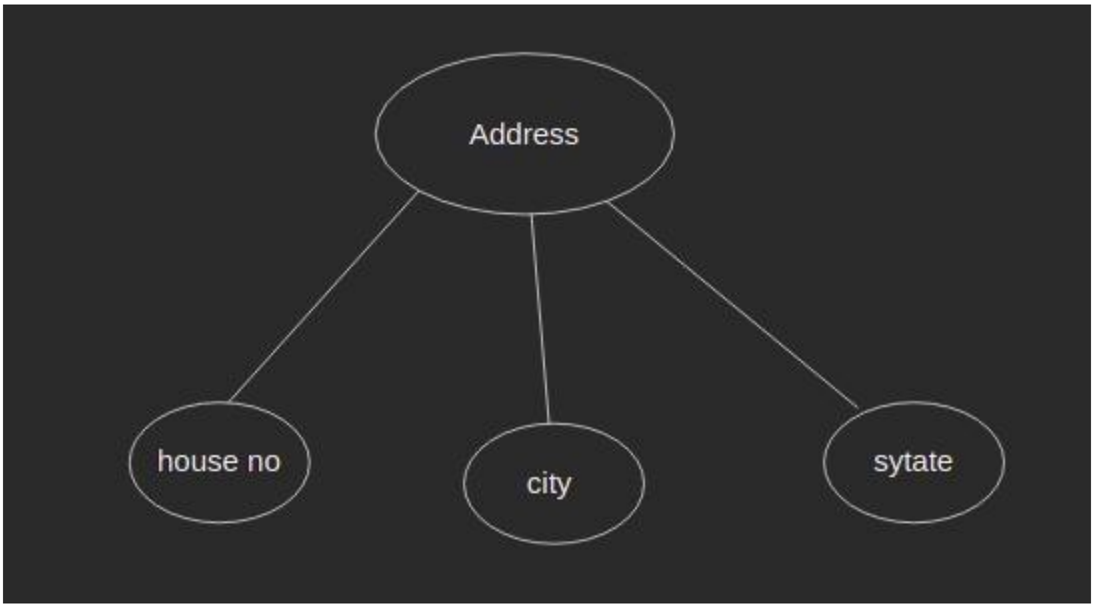
Key Attribute

The attribute which uniquely identifies each entity in the entity set is called key attribute. For example, Roll No will be unique for each student. In ER diagram, key attribute is represented by an oval with underlying lines.



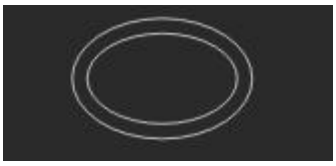
Composite Attribute –

An attribute composed of many other attributes is called as compositeattribute. For example, Address attribute of student Entity type consists of Street, City, State, and Country. In ER diagram, composite attribute is represented by an oval comprising of ovals.



Multivariate Attribute –

An attribute consisting more than one value for a given entity. For example, Phone No (can be more than one for a given student). In ER diagram, Multivariate attribute is represented by double oval.



Derived Attribute –

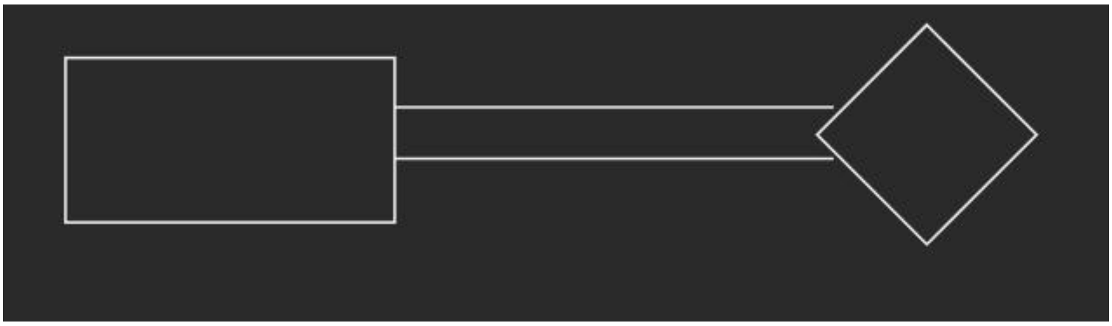
An attribute which can be derived from other attributes of the entitytype is known as derived attribute. e.g.; Age (can be derived Frodo). In ER diagram, derived attribute is represented by dashed oval.

Degree of a relationship set:

The number of different entities sets participating in a relationship set iscalled as degree of a relationship set.

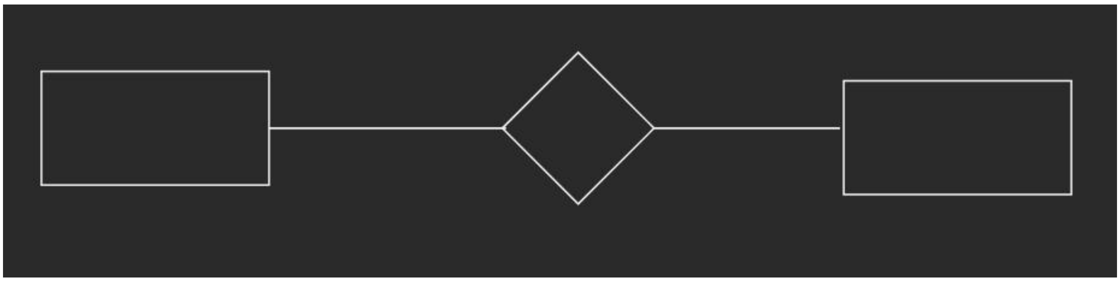
Unary Relationship –

When there is only ONE entity set participating in a relation, the relationship is called as unary relationship. For example, one person is married to only one person.



Binary Relationship –

When there are TWO entities set participating in a relation, the relationship is called as binary relationship. For example, Student is enrolled in Course.



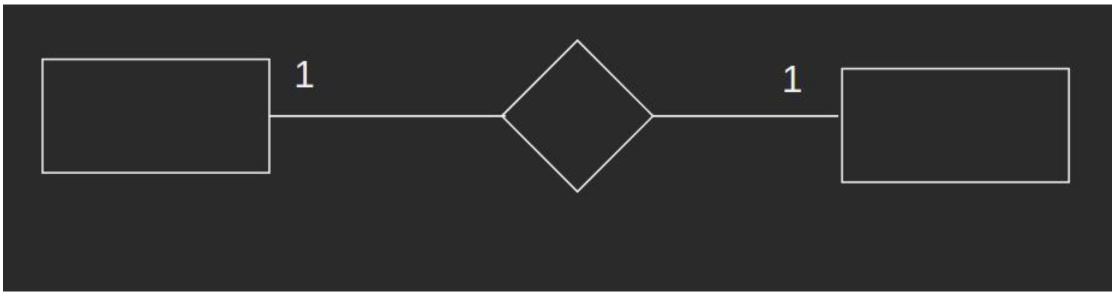
cardinal:

The number of times an entity of an entity set participates in a relationship set is known as cardinal.

cardinal can be of different types:

One to one

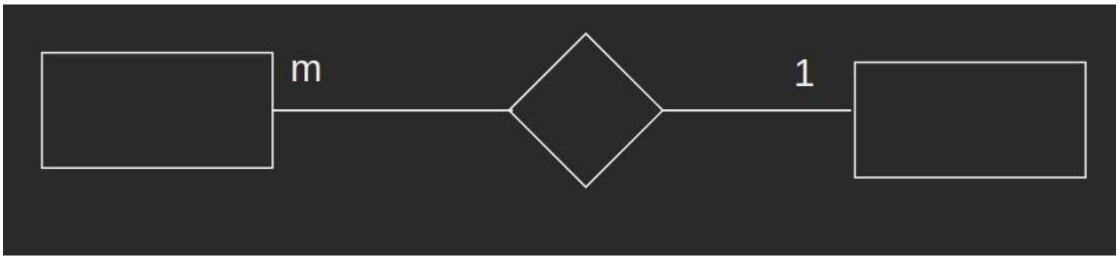
When each entity in each entity set can take part only once in the relationship, the cardinal is one to one. Let us assume that a male canmarry to one female and a female can marry to one male. So, the relationship will be one to one.



Many to one –

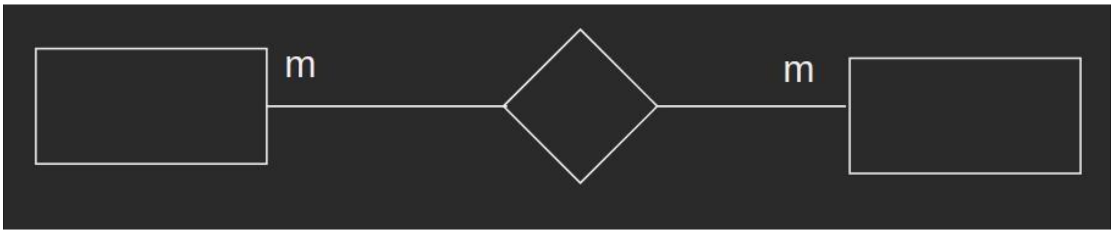
When entities in one entity set can take part only once in the relationshipset and entities in other entity set can take part more than once in the

relationship set; cardinal is many to one. Let us assume that a student can take only one course but one course can be taken by many students. So, the cardinal will be n to 1. It means that for one course there can be n students but for one student, there will be only one course.



Many to many –

When entities in all entity sets can take part more than once in the relationship cardinal is many to many. Let us assume that a student can take more than one course and one course can be taken by manystudents. So, the relationship will be many to many.



Participation Constraint:

Participation Constraint is applied on the entity participating in the relationship set.

Total Participation –

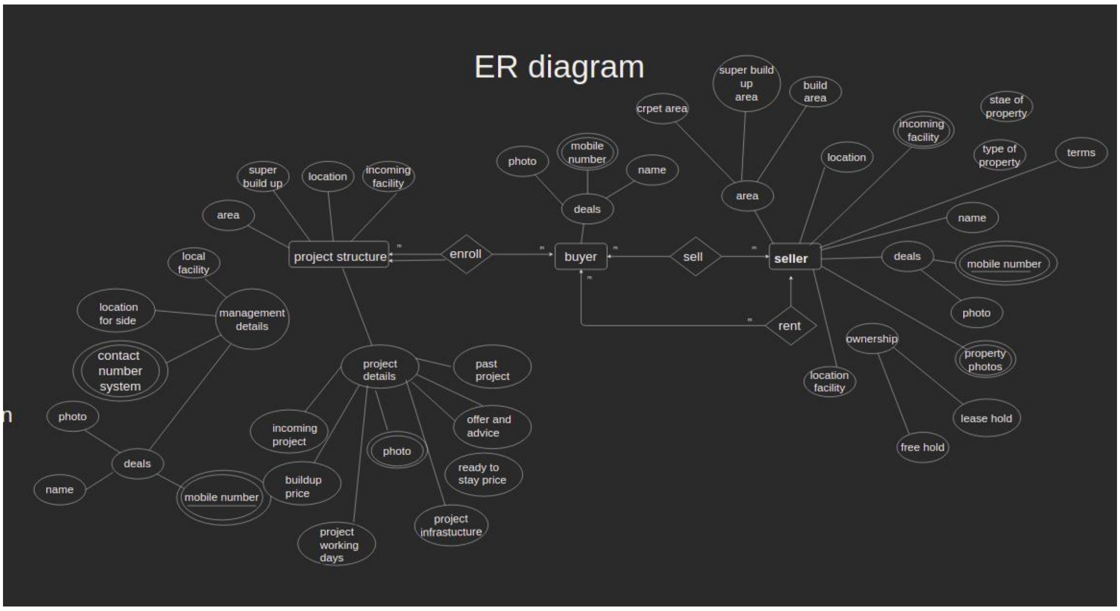
Each entity in the entity set must participate in the relationship. If each student must enroll in a course, the participation of student will be total. Total participation is shown by double line in ER diagram.

Partial Participation –

The entity in the entity set may or may NOT participate in the relationship. If some courses are not enrolled by any of the student, the participation of course will be partial.

The diagram depicts the 'Enrolled in' relationship set with Student Entity set having total participation and Course Entity set having partial participation.

ER diagram



ER diagram
project structure
enroll
buyer
sell
seller
rent
super build up
location
incoming facility
area
local facility
management details
location for side
contact number system
photo
deals
name
mobile number
project details
past project
incoming project
offer and advice
photo
buildup price
ready to stay price
project infrastucture
project working days
photo
mobile number
name
deals
crpet area
super build up area
build area
area
location
incoming facility
stae of property
type of property
terms
name
deals
mobile number
photo
ownership
property photos
location facility
lease hold
free hold

DFD: -

Basic DFD Diagram Symbols and Notations:

A data-flow diagram is a way of representing a flow of data through a process or a system (usually an information system). The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagram has no control flow, there are no decision rules and no loops. Specific operations based on the datacan be represented by a flowchart.

The flow of data of a system or a process is represented by DFD. Italso gives insight into the inputs and outputs of each entity and the process itself.

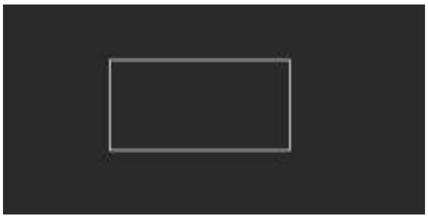
Level 0 that, also known as context diagrams, are the most basicdata flow diagrams. In a level 1 data flow diagram, the single process node from the context diagram is broken down into processed. As these processes are added, the diagram will need additional data flows and data stores to link them together.

DFD Level 0 is also called a Context Diagram. It’s a basic overview of the whole system or process being analysand or mudded. It’s designed to be an at-a-glance view, showing the system as a single high-level process, with its relationship to external entities.

The top level, is referred to Level 0 or the Context Diagram, it represents the system as one process box. Level 1 data flow diagramsshow incoming data flow, processes and output data flows. Level 1 Dads should show a process to handle each incoming data flow and aprocess to generate each output data flow.

Source or Sink:

Source or Sink is an external entity and acts as a source of systeminputs or sink of system outputs.



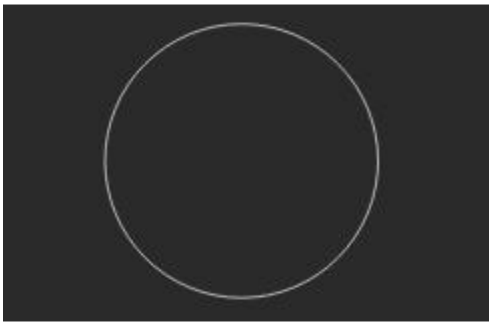
Data Store:

A set of parallel lines shows a place for the collection of data items. A data store indicates that the data is stored which can be used at a later stage or by the other processes in a different order.The data store can have an element or

group of elements.

Process

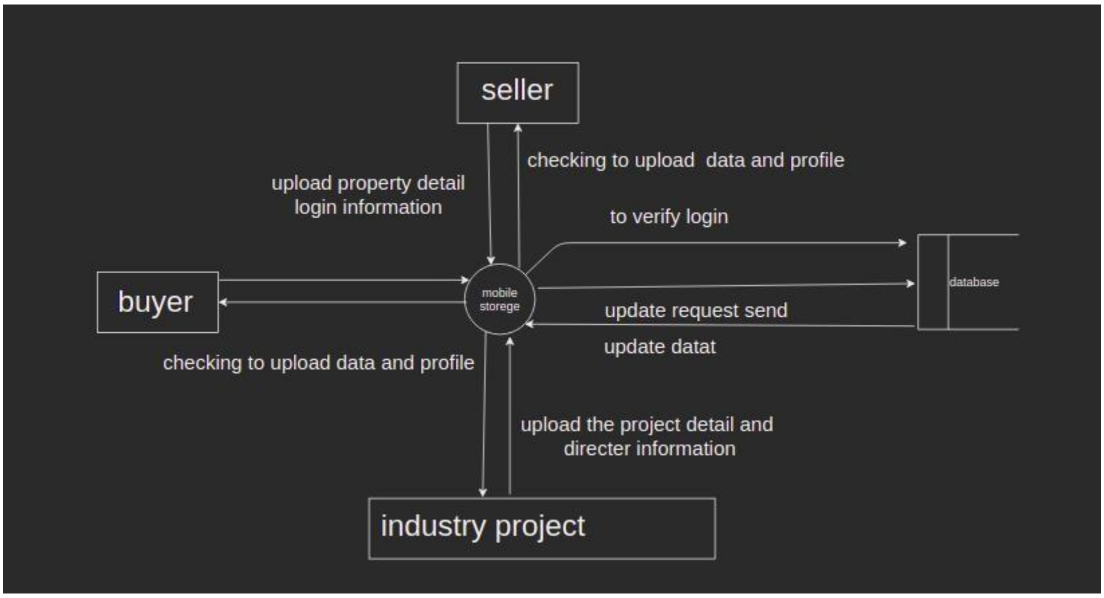
A data store does not generate any operations but simply holds data for later access. Data stores could consist of files held long termor a batch of documents stored briefly while they wait to be processed.



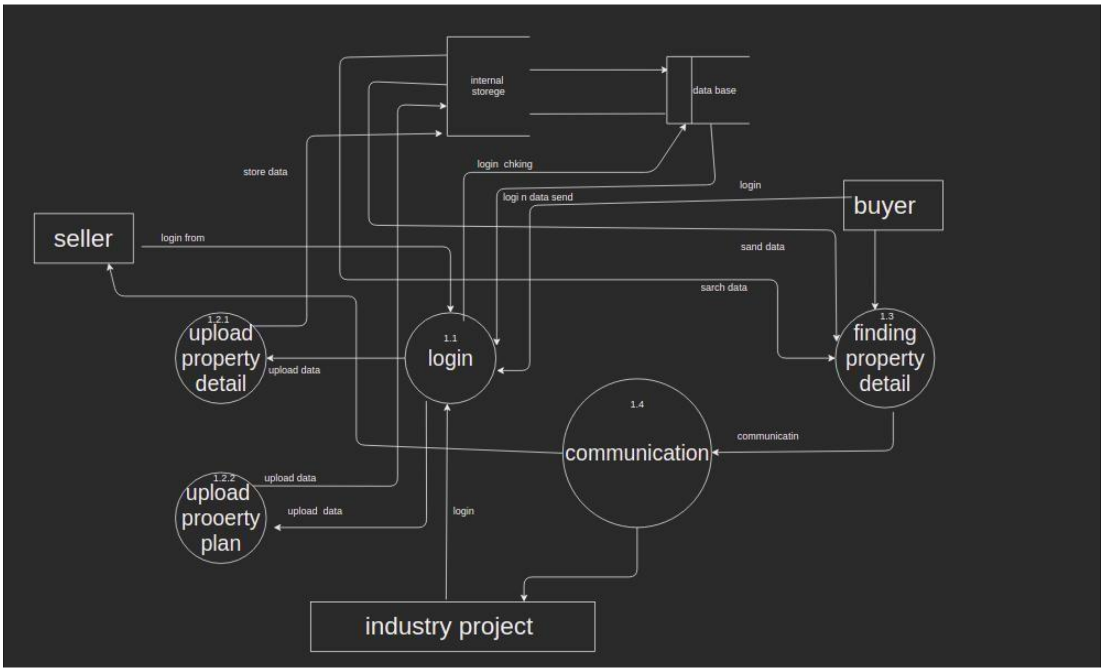
Data Flow:

Movement of data between external entities, processes and data stores is represented with an arrow symbol, which indicates the direction of flow.

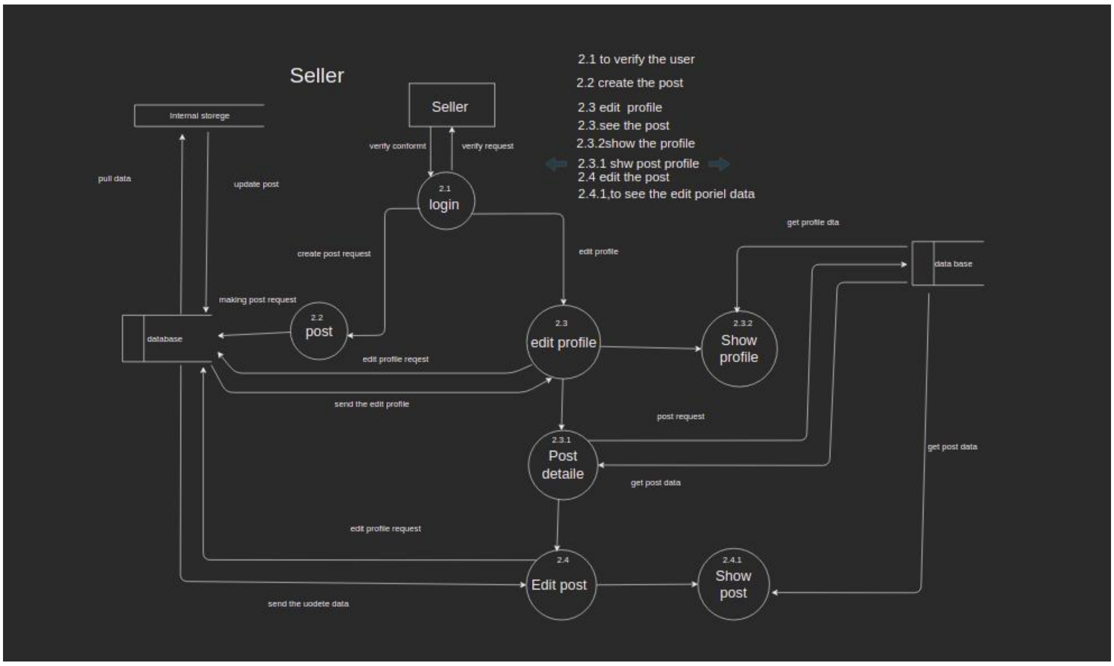
Context Leave Diagram


seller
upload property detail
login information
checking to upload data and profile
to verify login
buyer
mobile
storege
database
update request send
update datat
checking to upload data and profile
upload the project detail and
directer information
industry project

Level 1 diagram



2nd level diagram



Activity Diagram: -

Activity diagram is basically a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system.

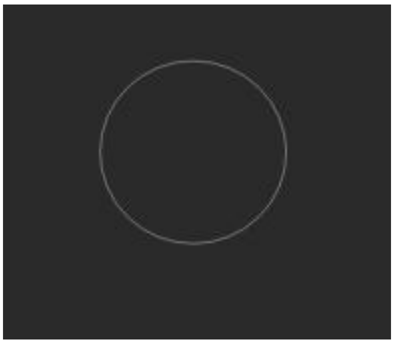
The control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent. Activity diagrams deal with all type of flow control by using different elements such as fork, join,etc.

The basic purposes of activity diagram are to capture the dynamic behavior of the system. Activity diagram is used to show message flow from one activity to another.

Basic Activity Diagram Symbols and Notations:

Initial State –

The starting state before an activity takes place is depicted using the initial state Represents the beginning of a process or work flow in an activity diagram. It can be used by itself or with a note symbol that explains the starting point.



Action or Activity State:

An activity represents execution of an action on objects or by objects. We represent an activity using a rectangle with rounded corners. Basically, any action or event that takes place is represented using an activity.

Connector Symbol:

Shows the directional flow, of the activity. An incoming arrow starts a step of an activity; once the step is completed, the flow continues with the outgoing arrow.

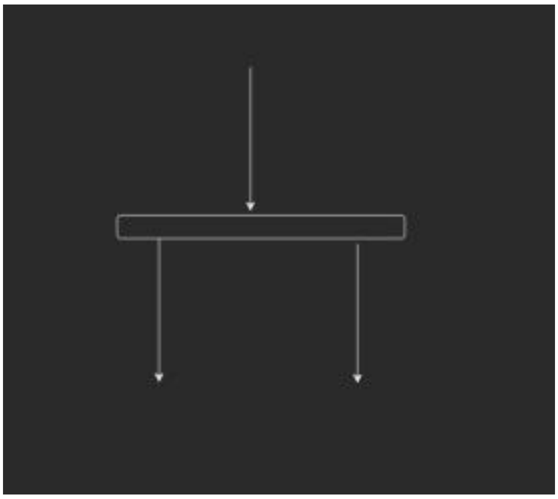
Joint Symbol:

Combines two concurrent activities and re-introduces them to a flow where only one activity occurs at a time. Represented with a thick vertical or horizontal line.

Fork Symbol:

Splits a single activity flow into two concurrent activities.

Symbolized with multiple Darrow lines from a join



Decision Symbol:

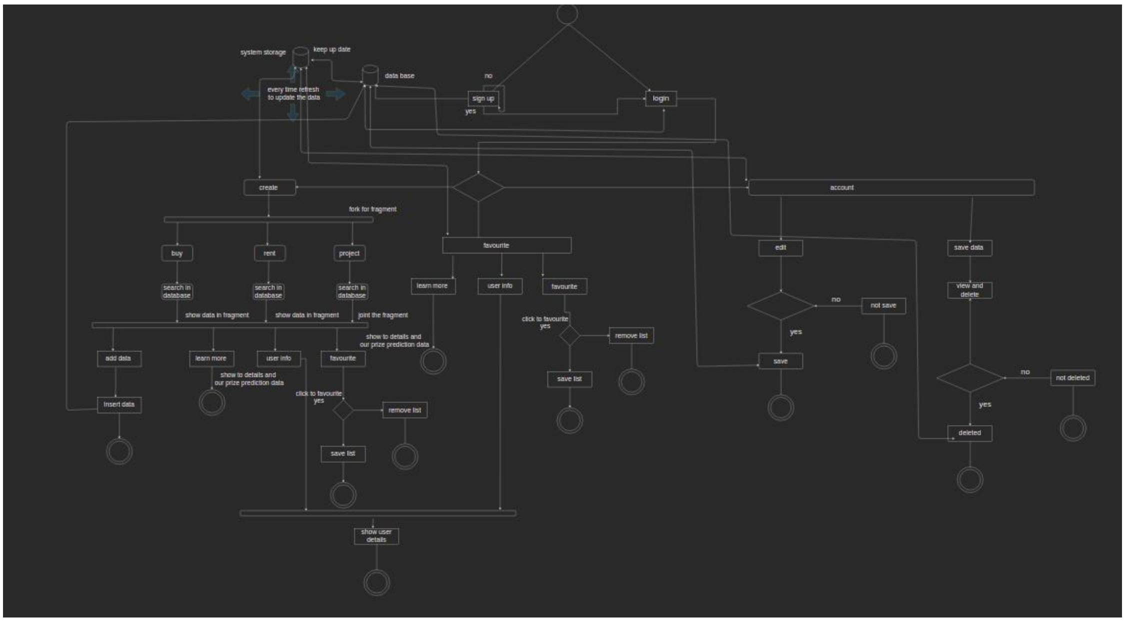
Represents a d e c i s i o n and always has at least two pathsBranching out with condition

text to allow users to view options.

End Symbol:

Marks the end state of an activity and represents the completion ofall flows of a process.

Activity diagram



system storage
keep up date
data base
every time refresh
to update the data
no
sign up
yes
login
create
fork for fragment
buy
rent
project
search in database
search in database
search in database
show data in fragment
show data in fragment
joint the fragment
add data
learn more
user info
favourite
insert data
show to details and
our prize prediction data
click to favourite
yes
remove list
save list
show to details and
our prize prediction data
favourite
learn more
user info
favourite
click to favourite
yes
remove list
save list
show user details
account
edit
no
not save
yes
save
save data
view and delete
no
not deleted
yes
deleted

Use case diagram: -

A use case diagram is a graphical depiction of a user's possible interactions with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. The actors are often shown as stick figures.

Use case diagrams are used to gather the requirements of a system including internal and external influences. These requirements are mostly design requirements. Hence, when a system is analysand to gather its functionalism, use cases are prepared and actors are identified.

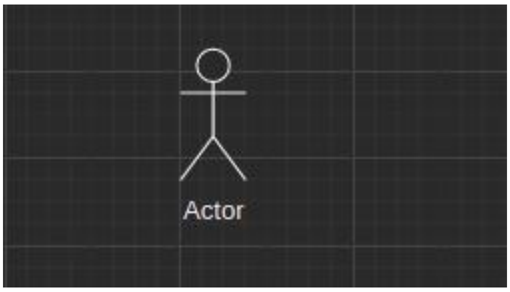
Basic Use-Case Diagram notations:

Use cases:

Horizontally shaped ovals that represent the different uses that auser might have.

Actors:

Stick figures that represent the people actually employing the usecases.

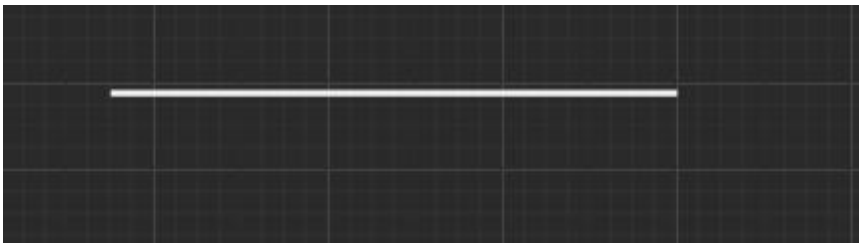


relationships:

A line between actors and use cases. In complex diagrams, it is important to know which actors are associated with which use cases.A relationship also displays how two use cases are related to each other.

COMMUNICATES :

The behavioral relationship communicates is used to connect an actor to a use case. Remember that the task of the use case is to givesome sort of result that is beneficial to the actor in the system. Therefore, it is important to document these relationships between actors and use cases. In our first example, a student communicates with Enroll in Course. Examples of some components of a student enrollment example are shown in the use case diagrams in the figurebelow



INCLUDES:

The includes relationship (also called uses relationship) describes the situation in which a use case contains behavior that is common to more than one use case. In other words, the common use case is included in the other use cases. A dotted arrow that points to the common use case indicates the includes relationship. An example would be a use case Pay Student Fees that is included in Enroll in Course and Arrange Housing, because in both cases students must pay their fees. This may be used by several use cases. The arrow points toward the common use case.

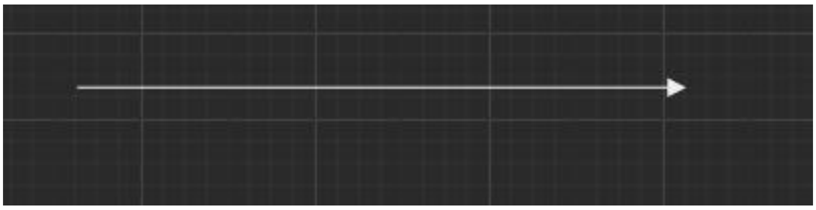
EXTENDS:

The extends relationship describes the situation in which one usecase possesses the behavior that allows the new use case to handle avariation or exception from the basic use case. For example, the extended use case Student Health Insurance extends the basic use case Pay Student Fees. The

arrow goes from the extended to the basic use case.

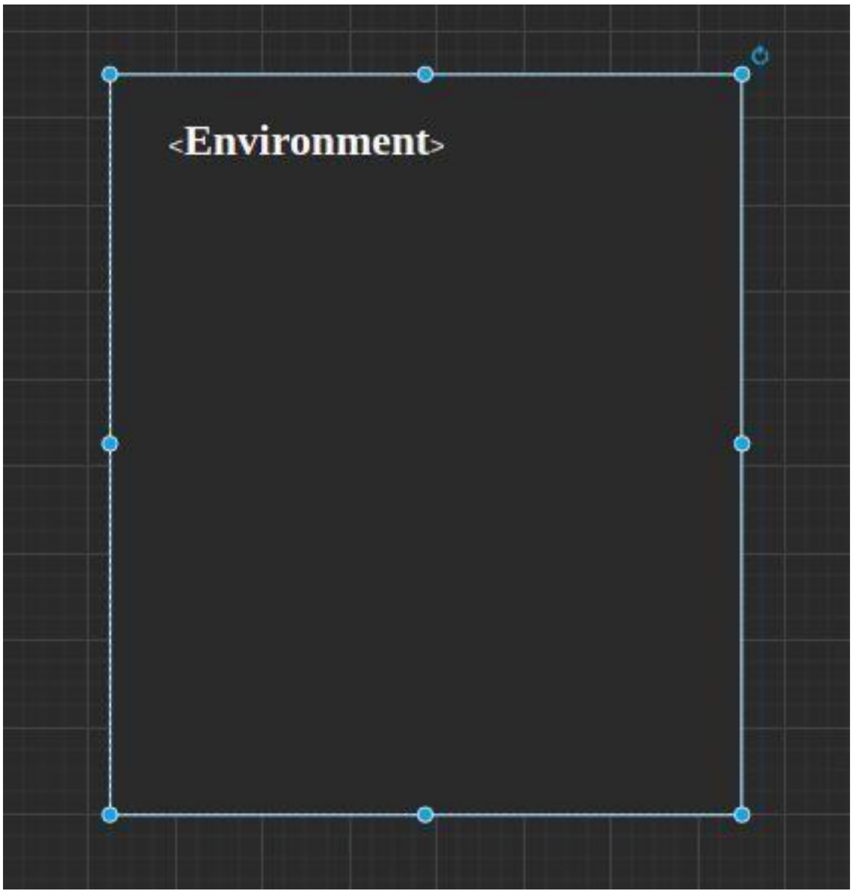
GENERALIZES:

The generalizes relationship implies that one thing is more typical than the other thing. This relationship may exist between two actors or two use cases. For example, a Part-Time Student generalizes a student. Similarly, some of the university employees are professors. The arrow points to the general thing.



Environment:

A box that sets a system scope to use cases.



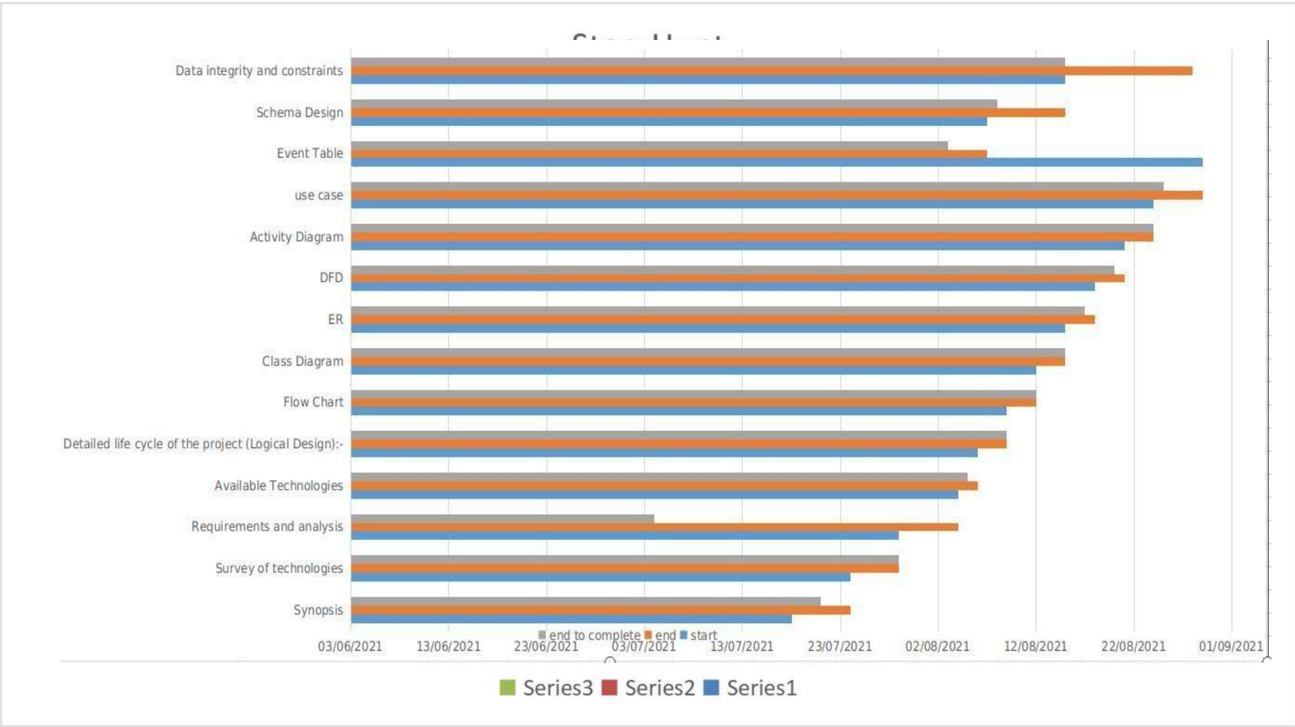
Event table: -

Event	Trigger	Source	Activity	Responses	destination
Register	Add user	user	Create account	New Account created	user
login	Verify user	user	login	Logged in	user

Seller post	Add post	user	Add property data	adding post	user
Project data insert	Add post	user	Add property data	Adding post	user
Buy	Visit property details	user	Visit collection	Visiting collection	user
logout	Sign out	user	Sign out	Sign out user	user

Gantt chart

sr no	Topic	Start	End	Date of Completion
1	Synopsis	20-07-2021	24-07-2021	23-07-2021
2	Survey of technologies	24-07-2021	29-07-2021	27-07-2021
3	Requirements and analysis	29-07-2021	04-08-2021	04-07-2021
4	Available Technologies	04-08-2021	06-08-2021	05-08-2021
4	Detailed life cycle of the project	06-08-2021	09-08-2021	09-08-2021
5	Flow Chart	09-08-2021	12-08-2021	12-08-2021
6	Class Diagram	12-08-2021	15-08-2021	15-08-2021
7	ER	15-08-2021	18-08-2021	17-08-2021
8	DFD	18-08-2021	21-08-2021	20-08-2021
9	Activity Diagram	21-08-2021	24-08-2021	24-08-2021
10	use case	24-08-2021	29-08-2021	25-08-2021
11	Event Table	29-08-2021	07-08-2021	03-08-2021
14	Schema Design	07-08-2021	15-08-2021	08-08-2021
15	Data integrity and constraints	15-08-2021	25-08-2021	22-08-2021
16	Code and Design	25-08-2021	01-01-2022	01-01-2022
17	Test cases	02-01-2022	18-01-2022	22-01-2022
18	Deployment	19-01-2022	02-02-2022	01-02-2022



Chapter 4

BASIC MODULE

A hierarchical database model is a data model in which the data are organized into a tree-like structure. The data are stored as records which are connected to one another through links. A record is a collection of fields, with each field containing only one value. The typeof a record defines which fields the record contains.

The hierarchical database model mandates that each child recordhas only one parent, whereas each parent record can have one or more child records. In order to retrieve data from a hierarchical database, the whole tree needs to be traversed starting from the root node.

Nosql format:

As explained in When to Use NoSQL Databases, NoSQL databases were developed during the Internet era in response to theinability of SQL databases to address the needs of web scale applications that handled huge amounts of data and traffic.

Companies are finding that they can apply NoSQL technology to agrowing list of use cases while saving money in comparison to operating a relational database. NoSQL databases invariably incorporate a flexible schema model and are designed to scale horizontally across many servers, which makes them appealing for large data volumes or application loads that exceed the capacity of asingle server **Schema Design:**

Generally speaking, because NoSQL databases are designed to store data that does not have a fixed structure that is specified prior to developing the physical model, developers focus on the physical data model. They are typically developing applications for massive, horizontally distributed environments. This puts an emphasis on figuring out how the availability and performance of the system will work. But they still need to think about the data model they will use toorganize

.

Document format

to Manama the all-information file format and use document and filecollection it also is NoSQL format to store a data

The popularity of NoSQL has been driven by the following **reasons**:

- The pace of development with NoSQL databases can be muchfaster than with a SQL database.
- The structure of many different forms of data is more easilyhandled and evolved with a NoSQL database.
- The amount of data in many applications cannot be served affordably by a SQL database.
- The scale of traffic and need for zero downtime cannot be handledby SQL.
- New application paradigms can be more easily supported.

Understanding Differences in the Four Types ofNoSQL

Databases

Document Databases:

A document database stores data in JSON, BSON, or XML documents (not Word documents or Google docs, of course). In a document database, documents can be nested. Particular elements can beindexed for faster querying.

Documents can be stored and retrieved in a form that is much closer to the data objects used in applications, which means less translationis required to use the data in an application. SQL data must often beassembled and disassembled when moving back and forth between applications and storage.

Document databases are popular with developers because they havethe flexibility to rework their document structures as needed to suit their application, shaping their data structures as their application requirements change over time. This flexibility speeds development because in effect data becomes like code and is under the control of developers. In SQL databases, intervention by database administrators may be required to change the structure of a database.

The most widely adopted document databases are usually implemented with a scale-out architecture, providing a clear path to availability of both data volumes and traffic.

- Key-Value Stores

The simplest type of NoSQL database is a key-value store. Everydata element in the database is stored as a key value pair consistingof an attribute name (or "key") and a value. In a sense, a key-value store is like a key

- Column-Oriented Databases

While a relational database stores data in rows and reads data row byrow, a column store is organized as a set of columns. This means thatwhen you want to run analytic on a small number of columns, you can read those columns directly without consuming memory with the unwanted data. Columns are often of the same type and benefit from more efficient compression, making reads even faster. Columnar databases can quickly aggregate the value of a given column (adding up the total sales for the year, for example). Use cases includeanalytic.

Unfortunately, there is no free lunch, which means that while columnar databases are great for analytic, the way in which they write data makes it very difficult for them to be strongly consistent aswrites of all the columns require multiple write events on disk. Relational databases don't suffer from this problem as row data is written contiguously to disk.

- Graph Databases

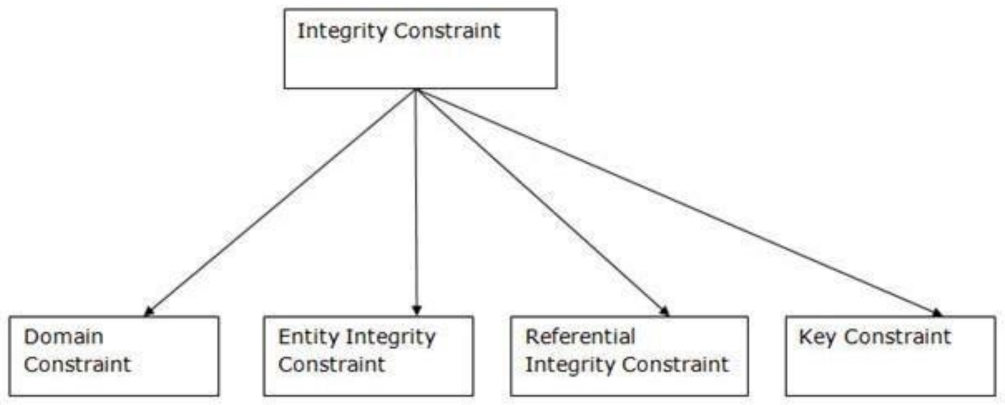
A graph database focuses on the relationship between data elements.Each element is stored as a node (such as a person in a social media graph). The connections between elements are called links or relationships. In a graph database, connections are firstclass elements of the database, stored directly. In relational databases, links are implied, using data to express the relationships.

A graph database is optimized to capture and search the connectionsbetween data elements, overcoming the overhead associated with Coining multiple tables in SQL.

Data integrity and constrains

Integrity Constraints

- Integrity constraints are a set of rules. It is used to maintain the quality of information.
- Integrity constraints ensure that the data insertion, updating, andother processes have to be performed in such a way that data integrity is not affected.
- Thus, integrity constraint is used to guard against accidentaldamage to the database.



Domain constraints

Domain constraints can be defined as the definition of a valid set of values for an attribute. The data type of domain includes string, character, integer, time, date, currency, etc. The value of the attribute must be available in thecorresponding domain.

Entity integrity constraints

The entity integrity constraint states that primary key value can't be null. This is because the primary key value is used to identify individual rows in relation and if the primary key has a null value, then we can't identify those rows. A table can contain a null value other than the primary key field.

Referential Integrity Constraints

A referential integrity constraint is specified between two tables. In the Referential integrity constraints, if a foreign key in Table 1 refers to the Primary Key of Table 2, then every value of the ForeignKey in Table 1 must be null or be available in Table 2.

Key constraints

Keys are the entity set that is used to identify an entity within its entity set uniquely. An entity set can have multiple keys, but out of which one key will be the primary key. A primary key can contain a unique and null value in the relational table.

Purpose of Normalization

Normalization is the process of structuring and handling the relationship between data to minimize redundancy in the relational table and avoid the unnecessary anomalies properties from the database like insertion, update and delete. It helps to divide large database tables into smaller tables and make a relationship betweenthem. It can remove the redundant data and ease to add, manipulateor delete table fields.

A normalization defines rules for the relational table as to whether it satisfies the normal form. A normal form is a process that evaluates each relation against defined criteria and removes the Multivariate, joins, functional and trivial dependency from a relation. If any data is updated, deleted or inserted, it does not cause any problem for database tables and help to improve the relational table' integrity andefficiency.

Objective of Normalization

It is used to remove the duplicate data and database anomalies fromthe relational table. Normalization helps to reduce redundancy and complexity byexamining new data types used in the table.

It is helpful to divide the large database table into smaller tables and link them using relationship. It avoids duplicate data or no repeating groups into a table.It reduces the chances for anomalies to occur in a database.

the types of anomalies that make the table inconsistency, loss of integrity, and redundant data.

Data redundancy

occurs in a relational database when two or more rows or columns have the same value or repetitive value leading to unnecessaryutilization of the memory.

Insert Anomaly:

An insert anomaly occurs in the relational database when some attributes or data items are to be inserted into the database without existence of other attributes. For example, In the Student table, if we want to insert a new course, we need to wait until the student enrolled in a course. In this way, it is difficult to insert new record inthe table. Hence, it is called insertion anomalies.

Update Anomalies:

The anomaly occurs when duplicate data is updated only in one place and not in all instances. Hence, it makes our data or table inconsistent state. For example, suppose there is a student 'James' who belongs to Student table. If we want to update the course in thestudent, we need to update the same in the course table; otherwise,the data can be inconsistent. And it reflects the changes in a table with updated values where some of them will not.

Delete Anomalies:

An anomaly occurs in a database table when some records are lost or deleted from the database table due to the deletion of other records. For example, if we want to remove Trent Bolt from the Student table, it also removes his address, course and other details from the student table. Therefore, we can say that deleting some attributes can remove other attributes of the database table.

Types of Normalization

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Bonce and C odd Normal Form (BCNF)

First Normal Form (1NF)

The table will be in First Normal Form (1NF) if all the attributes ofthe table contain only atomic values. We can also say that if a table holds the Multivariate data items in attributes or composite values, the relation cannot be in the first normal form. So, we need to make itfirst normal form by making the entries of the table atomic.

Second Normal Form (2NF):

A Relation will be in 2NF if it follows the following condition: The table or relation should be in 1NF or First Normal Form.

All the non-prime attributes should be fully functionally dependent onthe candidate key.

The table should not contain any partial dependency.

Third Normal Form (3NF):

The table will be in Third Normal Form (3NF) if it follows the given conditions:

- The table or relation should be in 2NF.
- It should not contain any transitive dependency. A Transitive Dependency is that any non-prime attribute determines or depends on the other non-prime attribute.
- A relation is in 3NF if FD X determines Y ('X' -> 'Y') satisfies one of thefollowing conditions:
- If X -> Y is a trivial FD, i.e., Y is a subset of X.
- If X -> Y, where X is a Super key.
- If X -> Y, (Y - X) is a prime attribute.

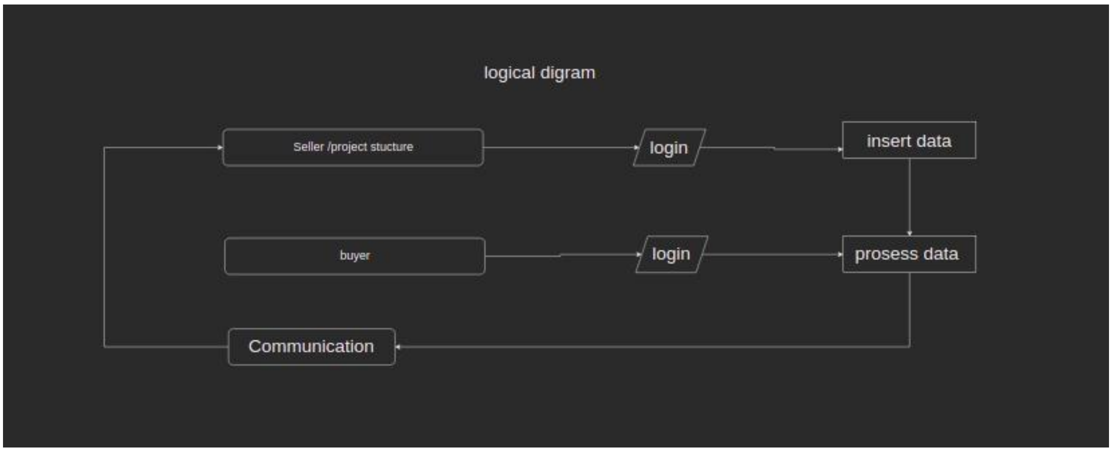
BCNF:

It stands for Bonce C odd Normal form, which is the next version of3NF. Sometimes, it is also pronounced as 3.5 NF. A normal form is said to be in BCNF if it follows the given conditions:

table or relation must be in 3NF.

If a relation R has functional dependencies (FD) and if A determines 2,where A is a super Key, the relation is in BCNF.

Logical diagram



SYSTEM WORKING

User (seller or project manager) will login and enter the property details. That data will be added in database and gets filtered out. Thatdetails will be visible to buyer so that buyer can compare the price, property structure and regarding features and then buyer can directlycommunicate the respective person.

Procedure design

Procedures are the rules, standards, methods design to increase the effectiveness of the information system. the procedures detail about the tasks to be performed in using the system. They serve as the ready recovers for the designers as well as for the users. sometime them perform the tasks of the supervisor over operators. In designing procedures, designers should

Understand the purpose and Quality standards of each procedure.Develop a step-bystep direction for each procedure. Document all the procedures.

chapter 5

5.1 coed detail and code efficiency

Login.java

```
package com.vm.brokfree.Authentication; import
android.content.Intent; import android.os.Bundle;
import android.util.Log; import
android.widget.TextView; import
android.widget.Toast; import
androidx.appcompat.app.AppCompatActivity; import
com.google.android.material.button.MaterialButton;
import
com.google.android.material.textfield.TextInputEditT
ext; import
com.google.firebase.firestore.CollectionReference;
import
com.google.firebase.firestore.FirebaseFirestore;
import com.google.firebase.firestore.Query; import
com.google.firebase.firestore.QueryDocumentSnapsh
ot; import com.vm.brokfree.R;
public class Login extends AppCompatActivity {

    TextInputEditText getUserPhoneNumber;
    MaterialButton login;
    TextView goToSignUp;

    FirebaseFirestore firebaseFirestore = FirebaseFirestore.getInstance();
    CollectionReference
    userNodeReference; String
    userEnterNumber; final
    private String TAG =
    "Login"; public boolean
    verificationFlag = true;
```

```
@Override              protected              void
onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_login); settingUpIds();
    initialTask(); login.setOnClickListener(v -> {
    toVerifyPhoneNumber();
    });
    goToSignUp.setOnClickListener(v -> goToSignUpPage());

}    private    void
settingUpIds() {
    getUserPhoneNumber =
    findViewById(R.id.user_phone_number); login =
    findViewById(R.id.login_btn); goToSignUp =
    findViewById(R.id.register);
}   private   void
initialTask() {



} private void toVerifyPhoneNumber() { userEnterNumber =
getUserPhoneNumber.getEditableText().toString().trim();
userNodeReference = firebaseFirestore
            .collection(getUserPhoneNumber.getEditableText().toString().tri
    m()); Query query = userNodeReference
            .whereEqualTo("phone_number",
getUserPhoneNumber.getEditableText().toString().trim());
//           firebaseFirestore
//                   .collection(getUserPhoneNumber.getEditableText().toString().trim())
//
          .whereEqualTo("phone_number",getUserPhoneNumber.getEditableText().toStrin
g().tri m())
//                   .get()
    query.get()
            .addOnCompleteListener(task -> { if
                (task.isSuccessful()) { for
```

```
                (QueryDocumentSnapshot document :
                task.getResult()) { if
                (document.getId().equals("user_info") &&
document.get("phone_number").equals(userEnterNumber)) {
                          verificationFlag = false;
                          Log.d(TAG, "onComplete: query find");
                   // Toast.makeText(getApplicationContext(), "you are user",
Toast.LENGTH_SHORT).show();
                          Intent intent = new Intent(Login.this, LoginOtp.class);
                          intent.putExtra("USER_NUMBER",
getUserPhoneNumber.getEditableText().toString().trim());
                          startActivity(in
                          tent); break;
                       }
                    }
                    if (verificationFlag) {
                Toast.makeText(getApplicationContext(), "not the user singUp",
Toast.LENGTH_SHORT).show();
                       Log.d(TAG,  "onComplete:  query
                    not find"); }
                } else {
  Toast.makeText(getApplicationContext(), "Error", Toast.LENGTH_SHORT).show();
                    Log.d(TAG,      "onComplete:     error
                occure query"); }
             });
   }
   private void goToSignUpPage() {
      Intent         intent         =         new         Intent(Login.this,         SignUp.class);
      intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
      startActivity(intent);
   }
```

5.2 TEST APPROACH Unit Testing

Unit testing is a software development process in which the smallest testable parts of an application, called units, are individually and independently scrutinized for proper operation. This testing methodology is done during the development process by the software developers and sometimes QA staff. The main objective of unit testing is to isolate written code to test and determine if it works as intended. Advantages and disadvantages of unit testing **Advantages to unit testing include:**

- The earlier a problem is identified, the fewer compound errors occur.
- Costs of fixing a problem early can quickly outweigh the cost of fixing it later.
- Debugging processes are made easier.
- Developers can quickly make changes to the code base.
- Developers can also re-use code, migrating it to new projects.

Integration testing

Integration testing is the second level of the software testing process comes after unit testing. In this testing, units or individual components of the software are tested in a group. The focus of the integration testing level is to expose defects at the time of interaction between integrated components or units.

Unit testing uses modules for testing purpose, and these modules are combined and tested in integration testing. The Software is developed with a number of software modules that are coded by different coders or programmers. The goal of integration testing is to check the correctness of communication among all the modules.

Advantages of Integration Testing

- Performing integration testing offers a lot of benefits. Some of them are listed below:

- It makes sure that integrated modules work properly as intended
- The tester can start testing once the modules to be tested are available
- It detects errors related to the interface between modules

- Helps modules interact with API’s and other third-party tools
- Typically covers a large volume of the system, so more efficient
- Increases the test coverage and improves the reliability of tests

Test case

A test case is a set of actions executed to verify a particular feature or functionality of your software application. A Test Case contains test steps, test data, precondition, postcondition developed for specific test scenario to verify any requirement. The test case includes specific variables or conditions, using which a testing engineer can compare expected and actual results to determine whether a software product is functioning ass per the requirements of the customer.

Chapter 6

6.1 Test reports and case table

Sr no	Action	input	Actual output	Expected out put	Test result
1	Log in	Phone no, otp	Login user	Login user	Pass
2	Sing up	First name, lastname, phone no, otp	Register user	Register user	Pass
3	Insert record	Property data	Inserting data in database	Insert data	Pass
4	Delete record	Remove data	Remove data in database	Remove data	Pass
5	Edit record	Changing data	Update data in database	Update data	Pass
6	Shuffling data	Keyword to find	To show different view of data	To shuffling data all data view not set	Failed
7	searching	Keyword to searching	Search bases of keyword	To show base of keyword	Pass

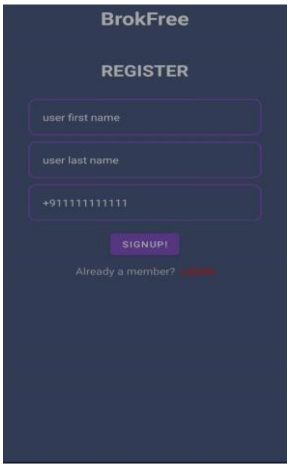
8	Predict price	Property related data	Predict price property	To predict price (57%) accuracy	Pass
9	Search with location	location	No data get	To get data shrouding are	Failed

Integrations Testing

Sr no	Action	input	Actual output	Expected out put	Test result
1	Login and signup	User first name, last name, phone number otp	register user and login user	User register and verify	pass

Test case 1: register user

Input data: Enter the phone number for verification and user first name and last name

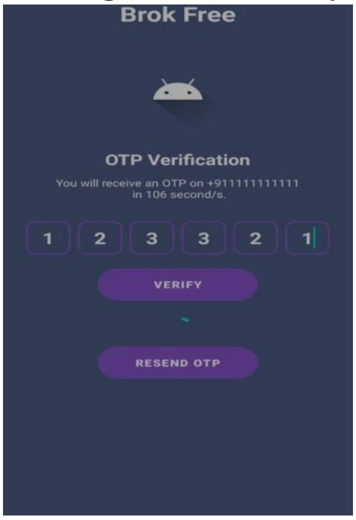


Result:

Expected Result:

add the user number, full name in database and register user and create profile and verify user number via otp using data base **Final result:**

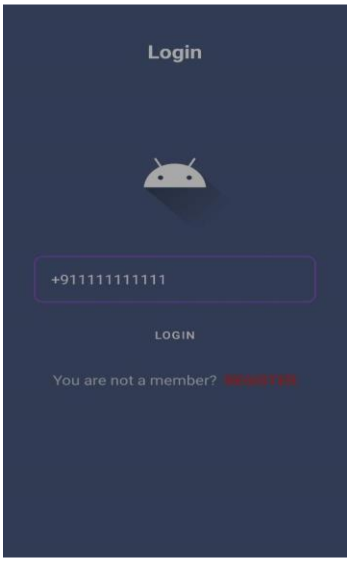
User register successfully



Identifier	Providers	Created ↓	Signed In	User UID
+911111111111		Feb 26, 2022	Mar 3, 2022	00pZOpbciTZvf8f6gJio65Quyv22

Case 2: user login

input: Enter the phone number for verification

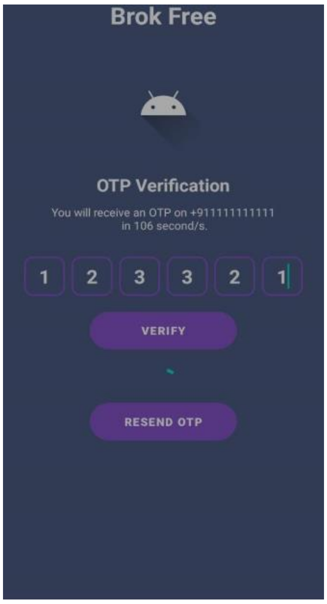


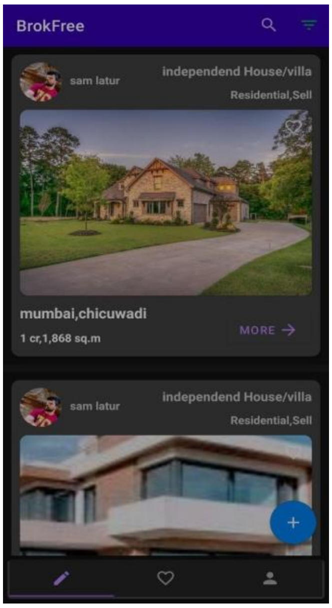
Result:

Expected Result:

verify the user number from database that register or not after verification user enter the otp the generated by database **Final result:**

User log in successfully

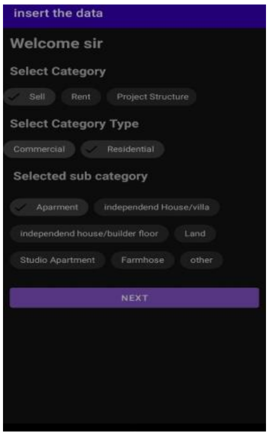


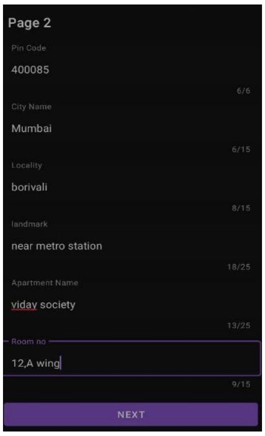


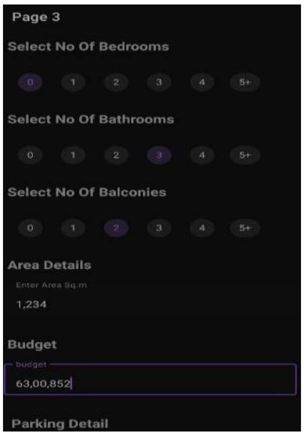
Test

Test case 3: Insert data

Input: User enter the property details







Result:

Expected Result:

Add user property data in database structure format

Final result:

user add successfully

case 4: delete record

Input: selected specific record to remove

Result:

Expected Result:

remove user selected property data in database structure format

Final result:

Recode remove successfully

Test

case 5: Edit record

Input: enter changing data to selected property

Result:

Expected Result: edit user selected property data in database structure format

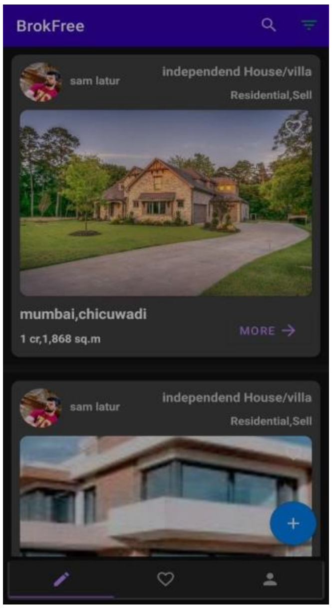
Final result: Recode edit successfully

6: shuffling data

refresh the lay out

Test case

Input:



Result:

Expected Result:

To show the different property data showing

Final result: shuffling data failed

No get data shuffling data then searching issues searching in google not get proper solution, I have referred stakoverflow.com, android.com git.com and I have found that my database doesn't support it.

7: searching data

searching keyword

Test case

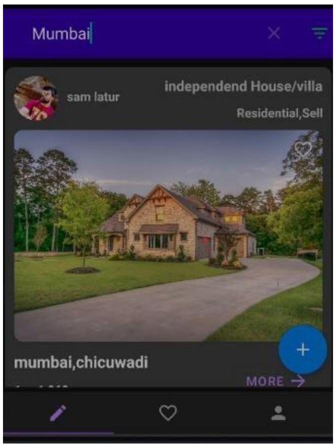
Input:

Result:

Expected Result:

User search data using city name as key word

Final result: Recode search successfully

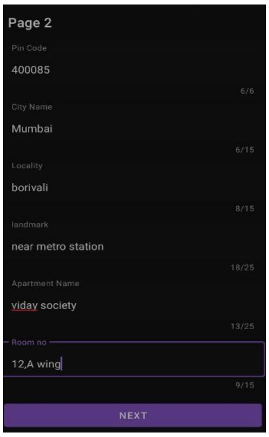


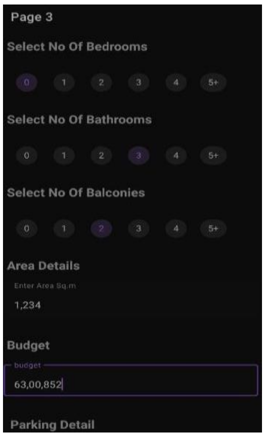
Test case

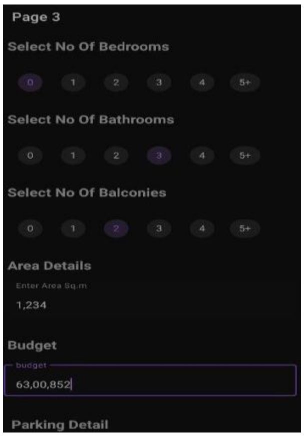
Input:

8: predict house price

user enter property data







Result:

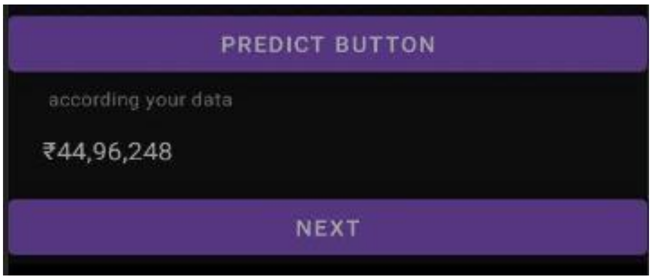
Expected Result:

Guess the price user property using inserting property data

Final result: predict value successfully

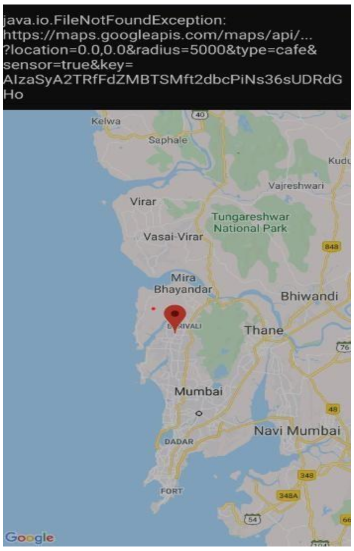
Test case

Input:



9: predict data using current location

use user current location, google map place api



Result:

Expected Result:

Test case

Input:

Guess the surrounding area user property and guess price also inserting property data

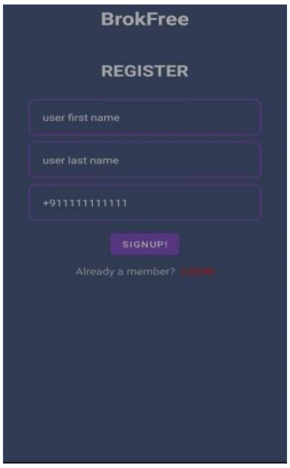
Final result: failed to collect information about the current location and information within 500m range

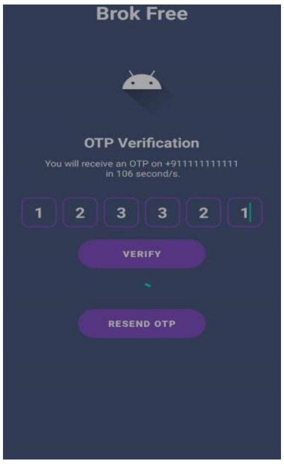
I have referred to google.com, stackoverflow.com, androidoc.com and get solution that api is paid for that feature and also this feature working as beta testing not compatible in android studio.

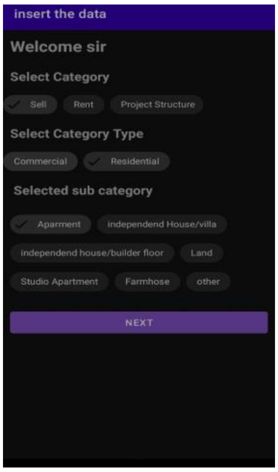
Integration Testing result

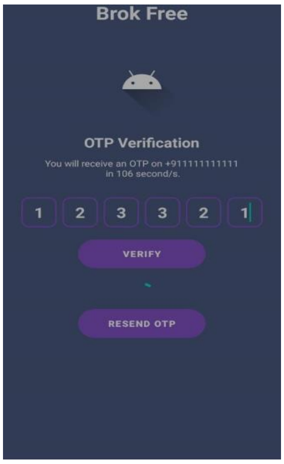
Test case 1: Login and signup Input: user first name,

user last name, phone no, otp







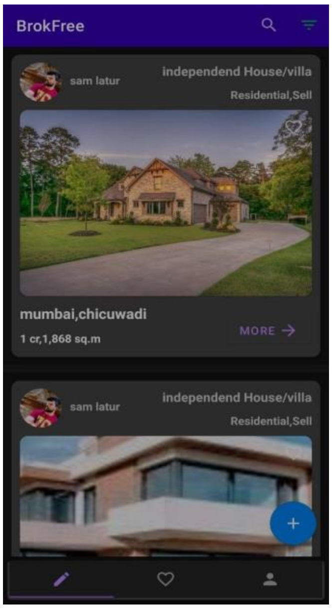


Result:

Expected Result:

To get register, verify and login user

Final result: user register and login successfully

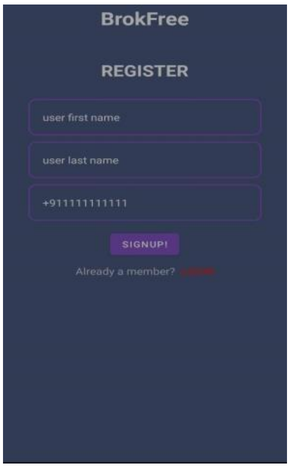


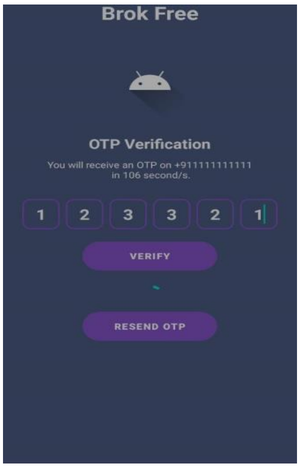
6.2 Test documentation

Login and sign up

User enter phone number, first name, last name to register yourself and verify the user phone number using otp and same follow login to enter the phone no and verify using otp

To verify user



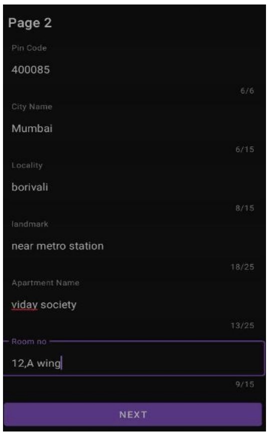


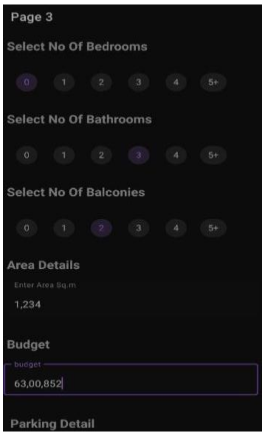
After entering otp firebase authenticates the user and store user information in database

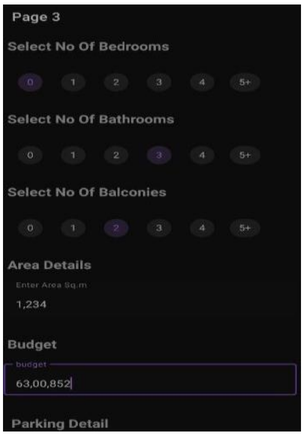
Identifier	Providers	Created	Signed In	User UID
+911111111111		Feb 26, 2022	Mar 3, 2022	00pZOpbciTZvf8f6gJio65Quyv22

Add property details

Add property details using add button you can add property. Select category, sub category, type of property, then add your address, select your property detail press next button to store to add property in data base







Predict features

As per the user entered information about their property AI (machine learning) predicts the estimated value of their property with 57.87% of accuracy

Edit and delete

User can modify and delete a property details their modification user can only modify address only the mention proviso user can delete their property details easily

User updated data can update modify data in database

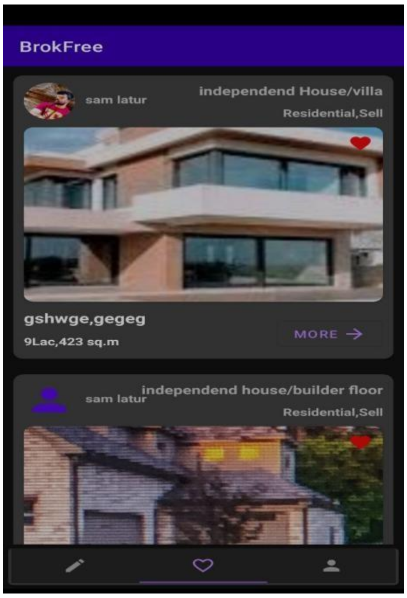
Searching data

User can search the property using location (city name, locality, pine code) to enter the searching keyword that search data in data base give appropriate result

When search completed the give result the searched by user

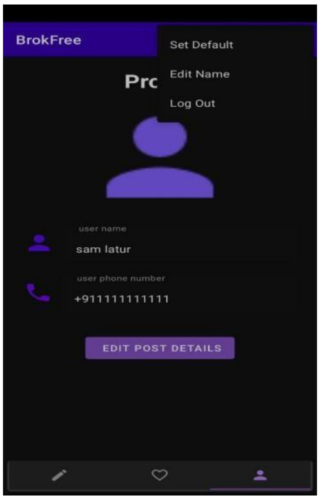
Add favorite

User can add property detail in favorite list and store the favorite list that show in second tap to easy remember other property data



Profile view

User can see the profile picture and add profile picture, edit name log out, User have access to see profile details. Also update the details



Chapter 7

7.1 Conclusion and Future Work

BrokFree android application is a platform that helps to buy sell and rent your property. It gives a platform that without give any broker charge and help the sell, buy and rent your property. It has many features that help to best deal for your property.

It helps to classify diffident category for property and manage the best deal like commercial, residential, flat, bang-lo, society etc. So Much Category help to manage system very well manner and you well ableto buy slum area.in this system work on only seller and buyer no onecan involve that system buyer can communicate direct to seller thorough this application and you safely communicant withbuyer/seller without any broker.

- To improve data location base Data, get system and to get predict model base increasing accuracy
- Get location api support to improve the getting perfect address using google map
- To improve customizable search to get more accurate result

REFERENCE

https://www.android.com/docs

https://www.github.com

https://www.javatpoint.com

https://www.flask.com

https://www.youtube.com

https://www.w3Schools.com

https://firebase.google.com